National University of Computer and Emerging Sciences
Islamabad Campus

CS3001

COMPUTER NETWORKS

FINAL PROJECT

REPORT

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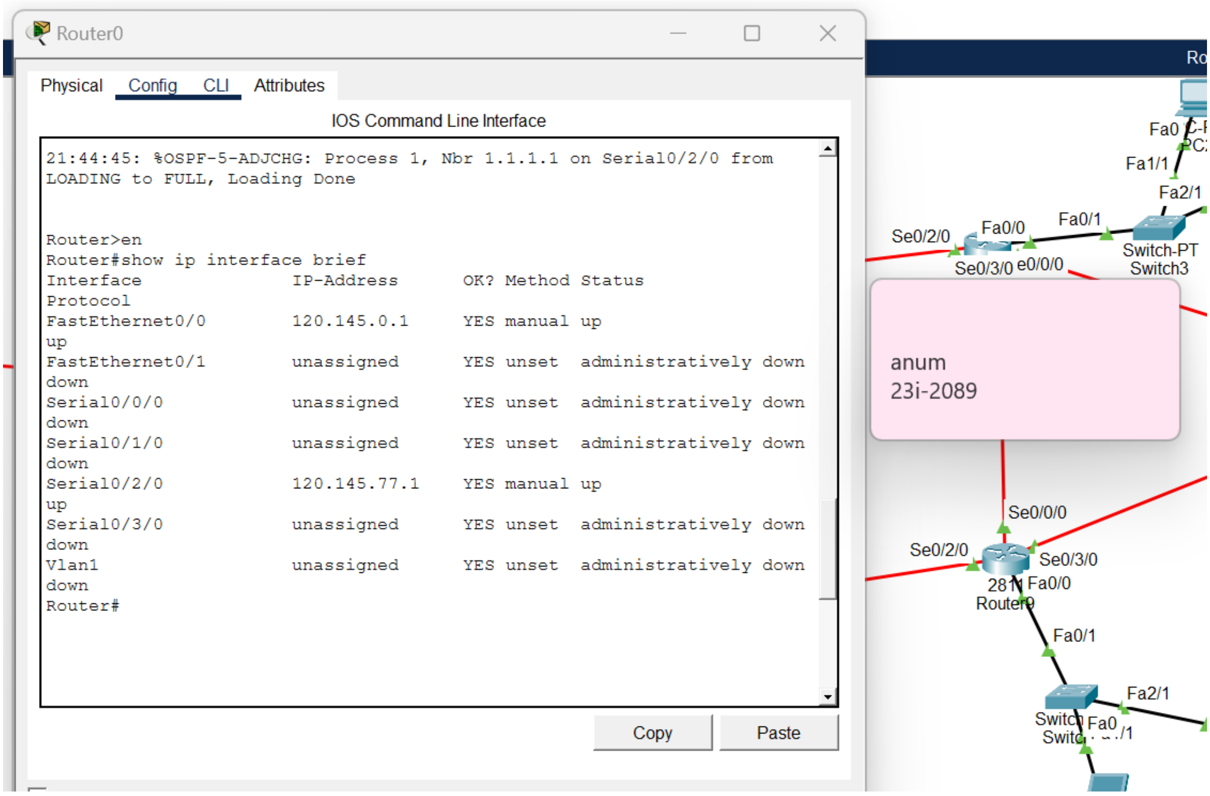
Submission Date: 7th December, 2025

1. Basic IP / Routing per Block

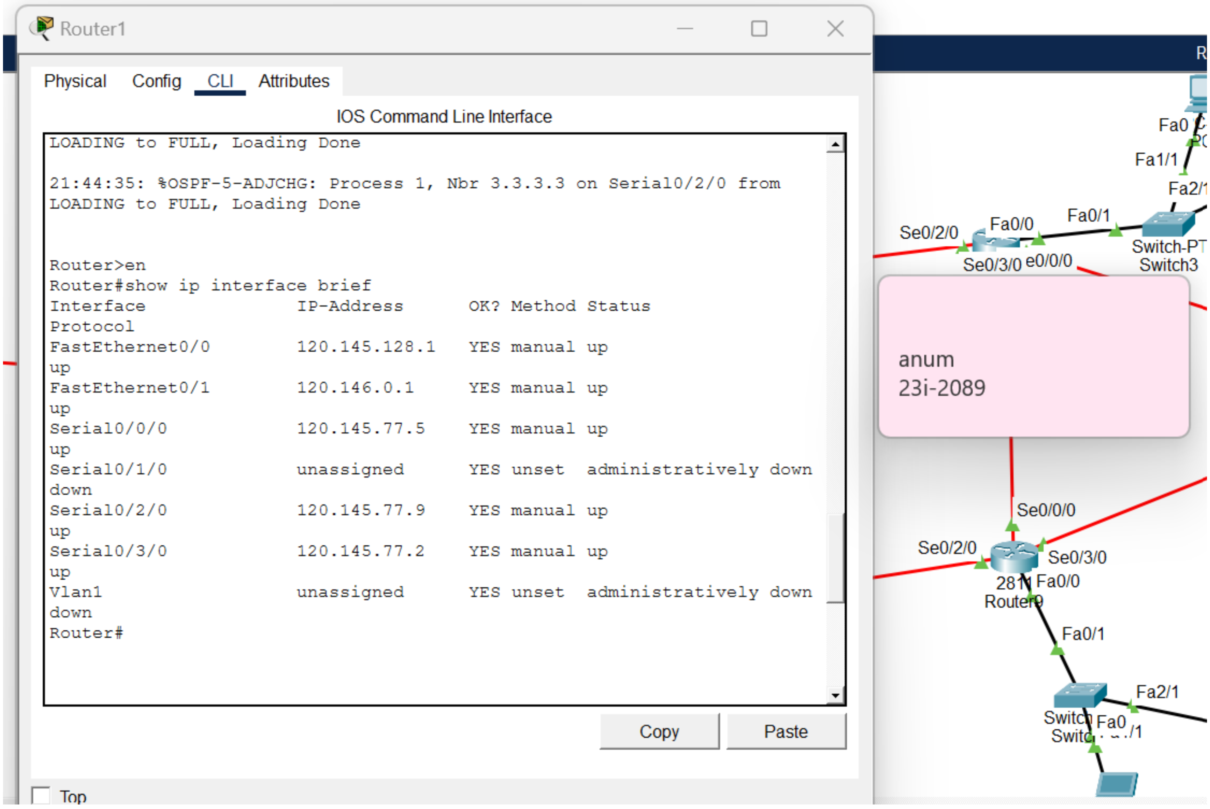
1.1. Show router IPs & masks (VLSM proof)

By **show ip interface brief** on Router0, Router1 and Router8, usage of /24 and /27 and /30 is showed.

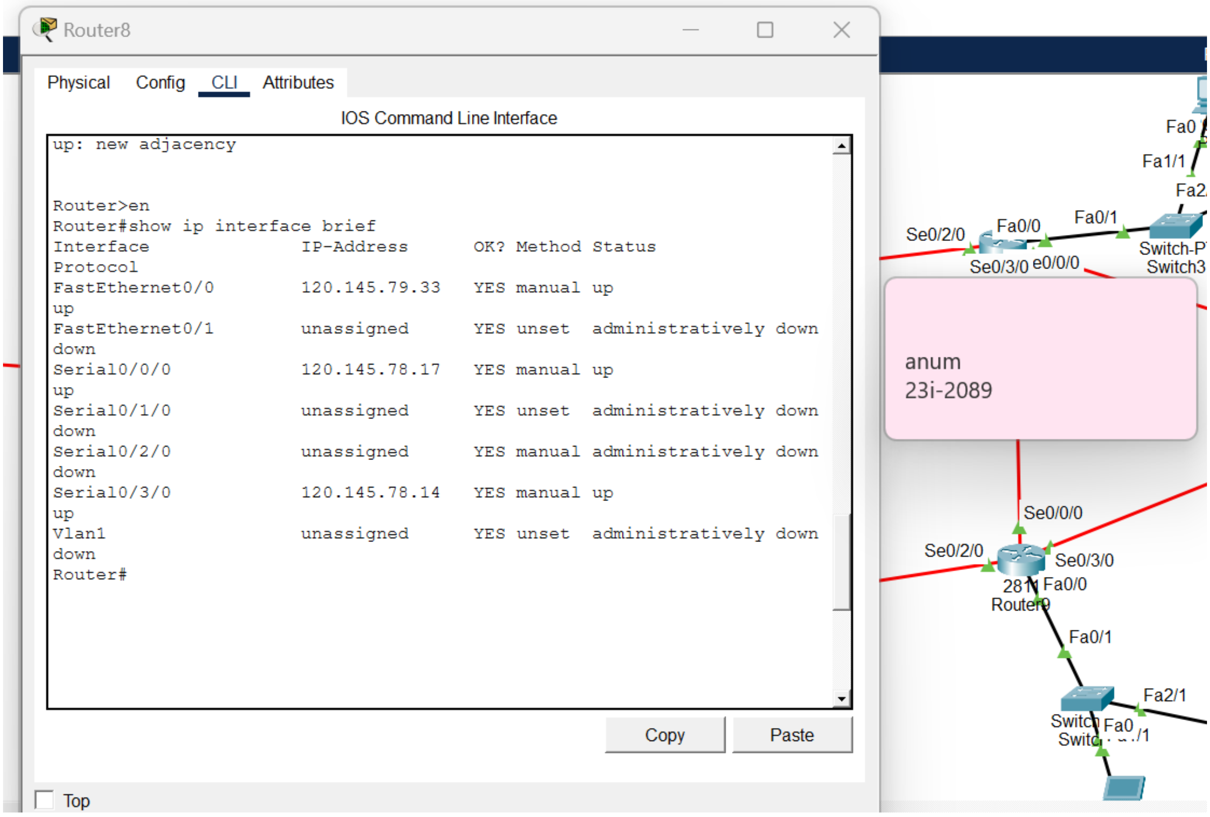
On router0:



On router1:



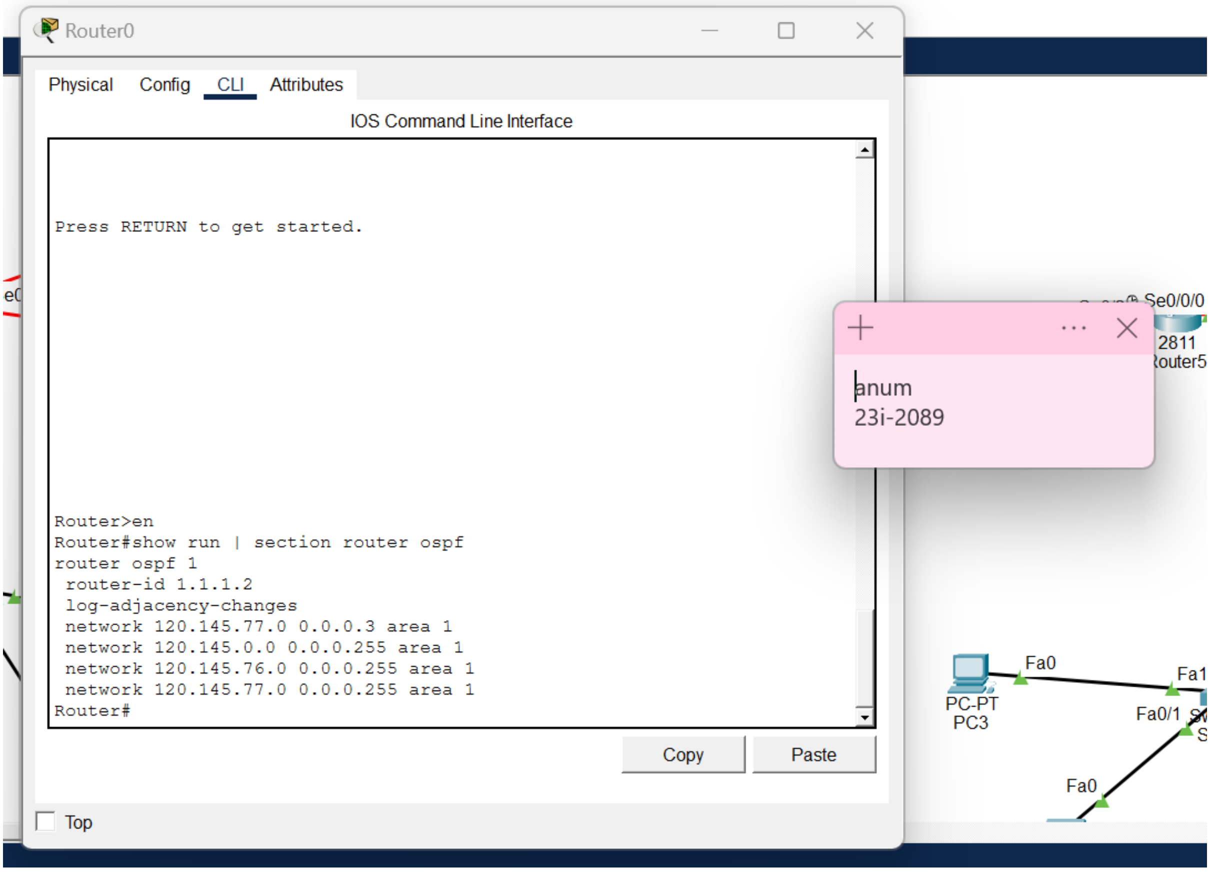
On Router8



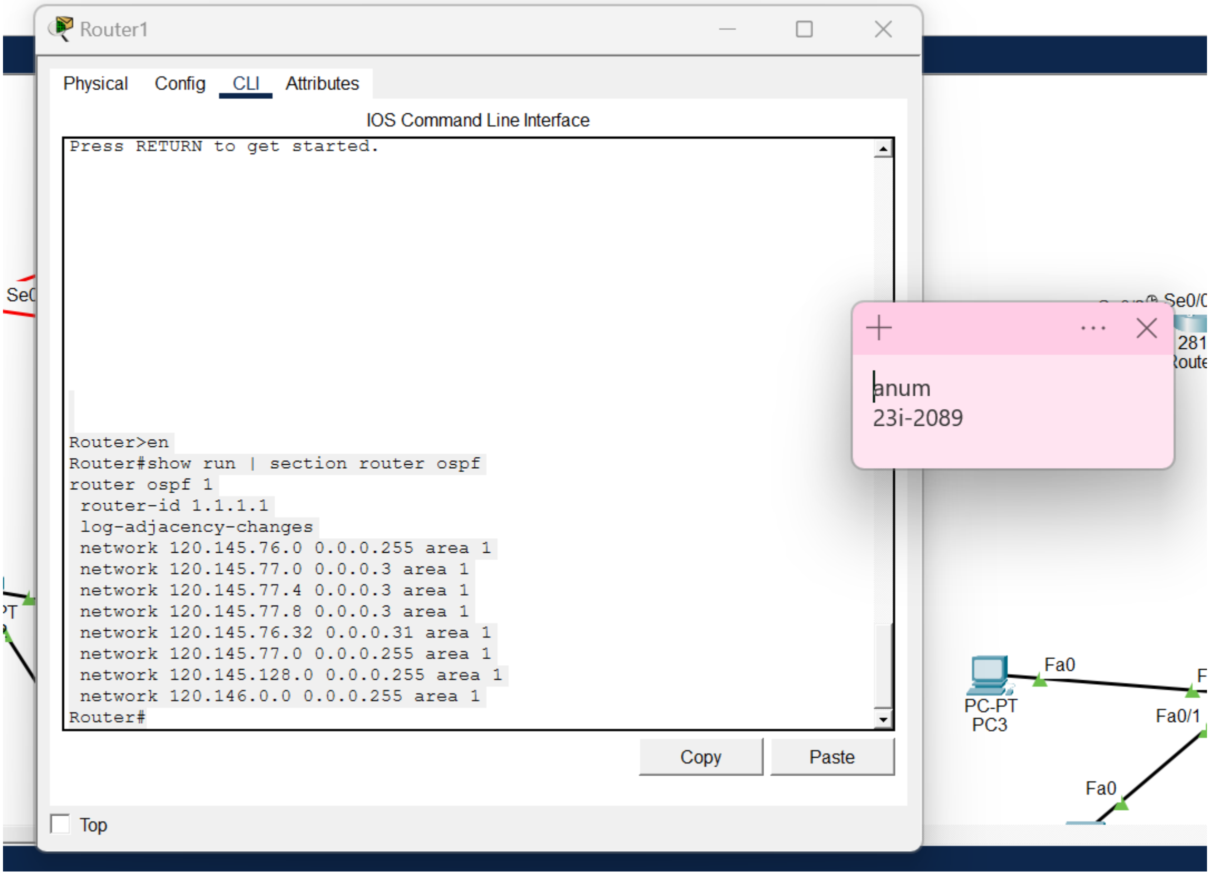
1.2. Correct Routing Protocol on OSPF, EIGRP and RIP

A. On routers of Area1/ OSPF1.

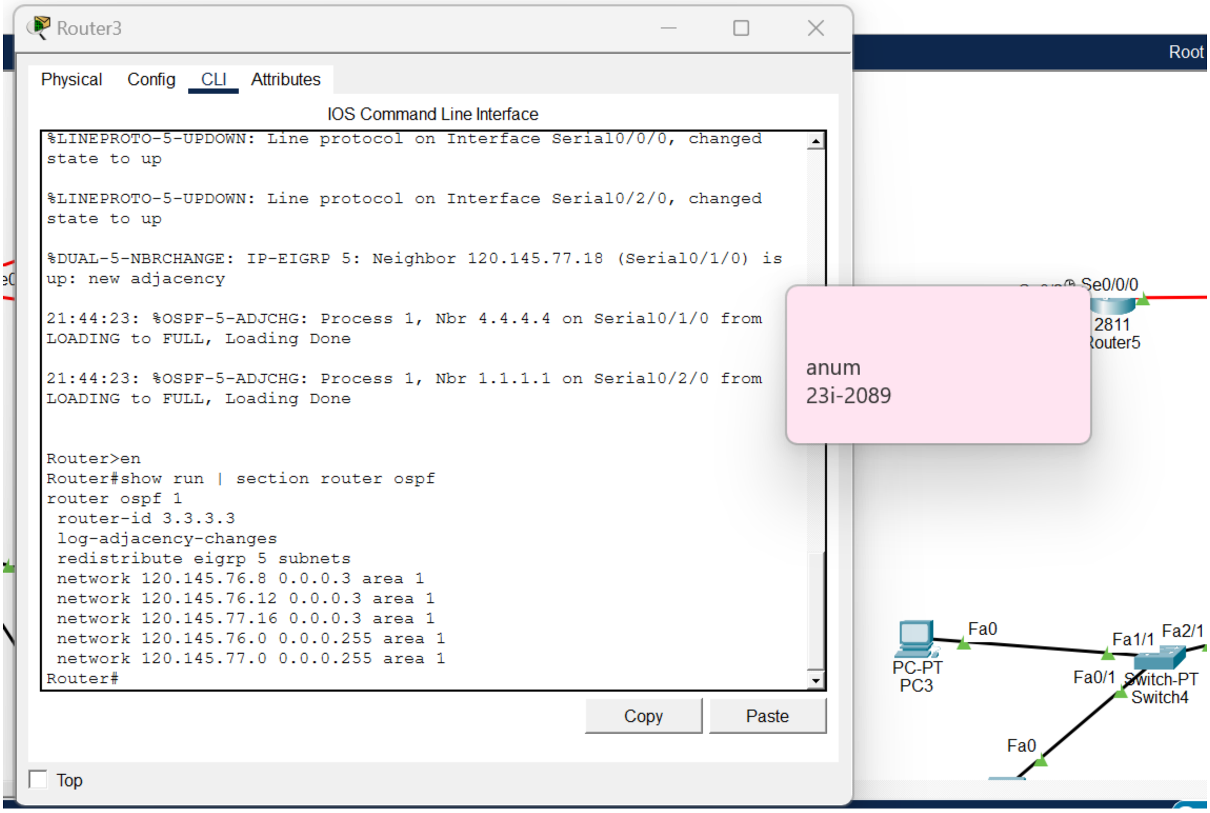
Router0:



On Router1:


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
Router>en
Router#show run | section router ospf
router ospf 1
router-id 1.1.1.1
log-adjacency-changes
network 120.145.76.0 0.0.0.255 area 1
network 120.145.77.0 0.0.0.3 area 1
network 120.145.77.4 0.0.0.3 area 1
network 120.145.77.8 0.0.0.3 area 1
network 120.145.76.32 0.0.0.31 area 1
network 120.145.77.0 0.0.0.255 area 1
network 120.145.128.0 0.0.0.255 area 1
network 120.146.0.0 0.0.0.255 area 1
Router#
Copy
Paste
Top
anum
23i-2089
PC-PT
PC3
Fa0
Fa0/1
Fa0

On Router3:


Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up
%DUAL-5-NBRCHANGE: IP-EIGRP 5: Neighbor 120.145.77.18 (Serial0/1/0) is up: new adjacency
21:44:23: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Serial0/1/0 from LOADING to FULL, Loading Done
21:44:23: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/2/0 from LOADING to FULL, Loading Done
Router>en
Router#show run | section router ospf
router ospf 1
router-id 3.3.3.3
log-adjacency-changes
redistribute eigrp 5 subnets
network 120.145.76.8 0.0.0.3 area 1
network 120.145.76.12 0.0.0.3 area 1
network 120.145.77.16 0.0.0.3 area 1
network 120.145.76.0 0.0.0.255 area 1
network 120.145.77.0 0.0.0.255 area 1
Router#
Copy
Paste
Top
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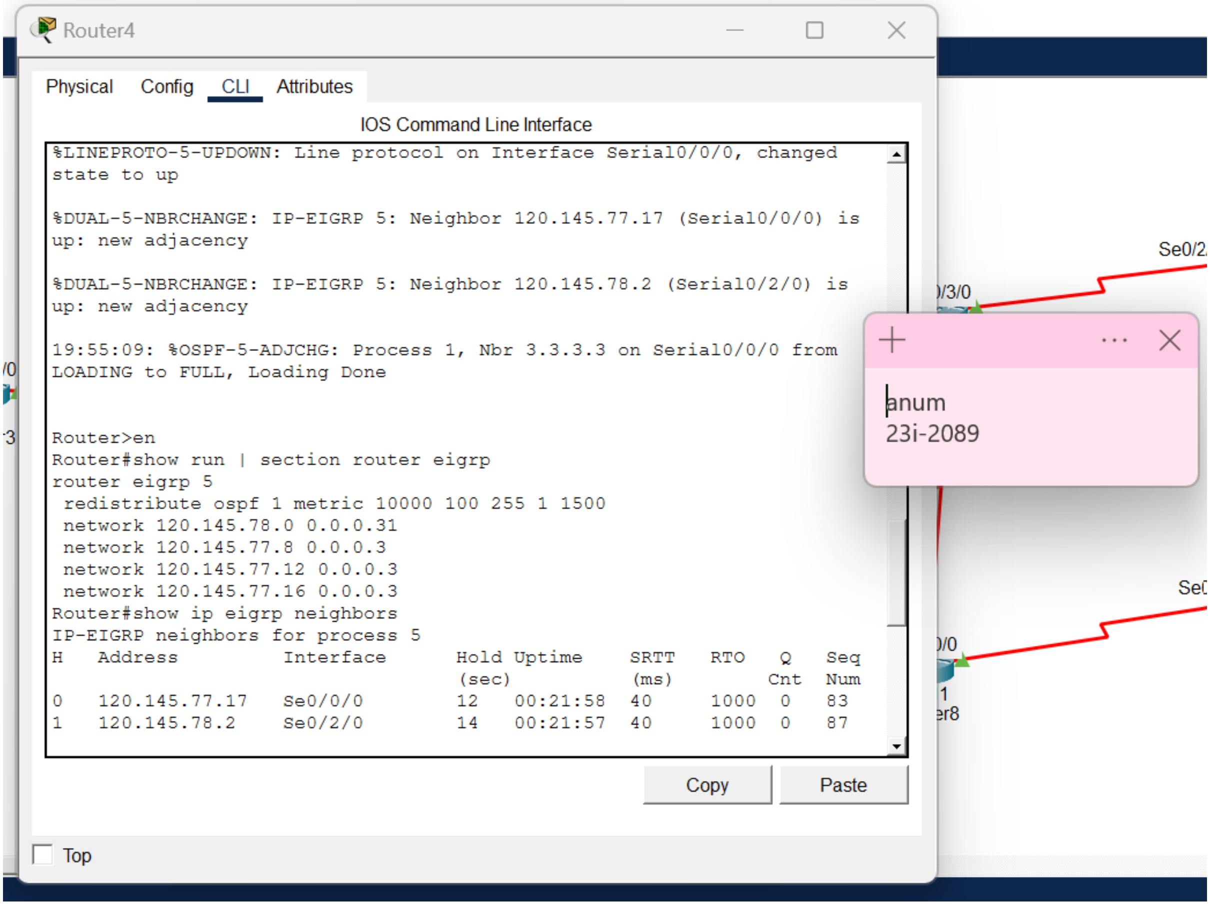
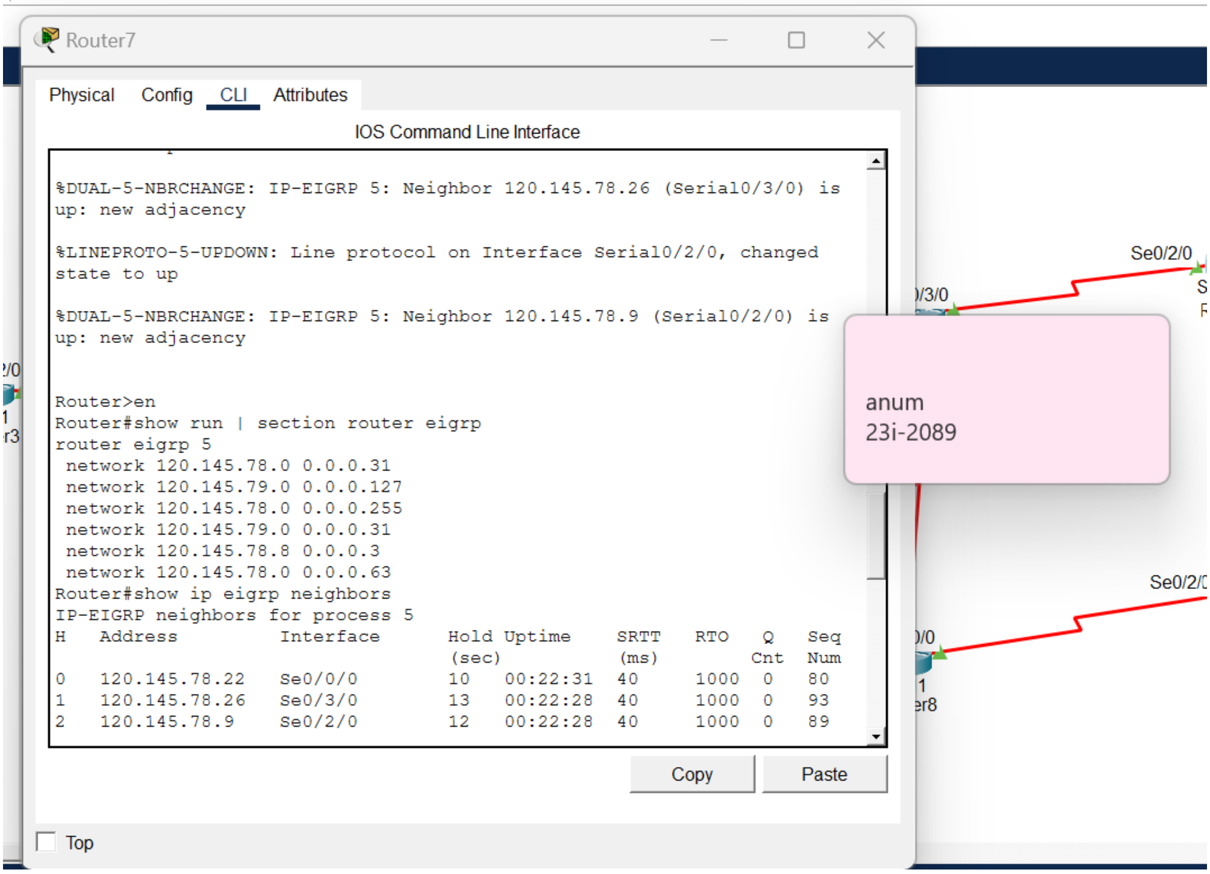
Root
Se0/0/0
2811
Fa0
PC-PT
PC3
Fa1/1
Fa2/1
Fa0/1
Switch-PT
Switch4
Fa0

B. On Routers of Pure EIGRP5 Block

On Router4:



Router7:


Router7
Physical
Config
CLI
Attributes
IOS Command Line Interface
%DUAL-5-NBRCHANGE: IP-EIGRP 5: Neighbor 120.145.78.26 (Serial0/3/0) is
up: new adjacency
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed
state to up
%DUAL-5-NBRCHANGE: IP-EIGRP 5: Neighbor 120.145.78.9 (Serial0/2/0) is
up: new adjacency
Router>en
Router#show run | section router eigrp
router eigrp 5
network 120.145.78.0 0.0.0.31
network 120.145.79.0 0.0.0.127
network 120.145.78.0 0.0.0.255
network 120.145.79.0 0.0.0.31
network 120.145.78.8 0.0.0.3
network 120.145.78.0 0.0.0.63
Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 5
H Address Interface Hold Uptime SRTT RTO Q Seq
(sec) (ms) Cnt Num
0 120.145.78.22 Se0/0/0 10 00:22:31 40 1000 0 80
1 120.145.78.26 Se0/3/0 13 00:22:28 40 1000 0 93
2 120.145.78.9 Se0/2/0 12 00:22:28 40 1000 0 89
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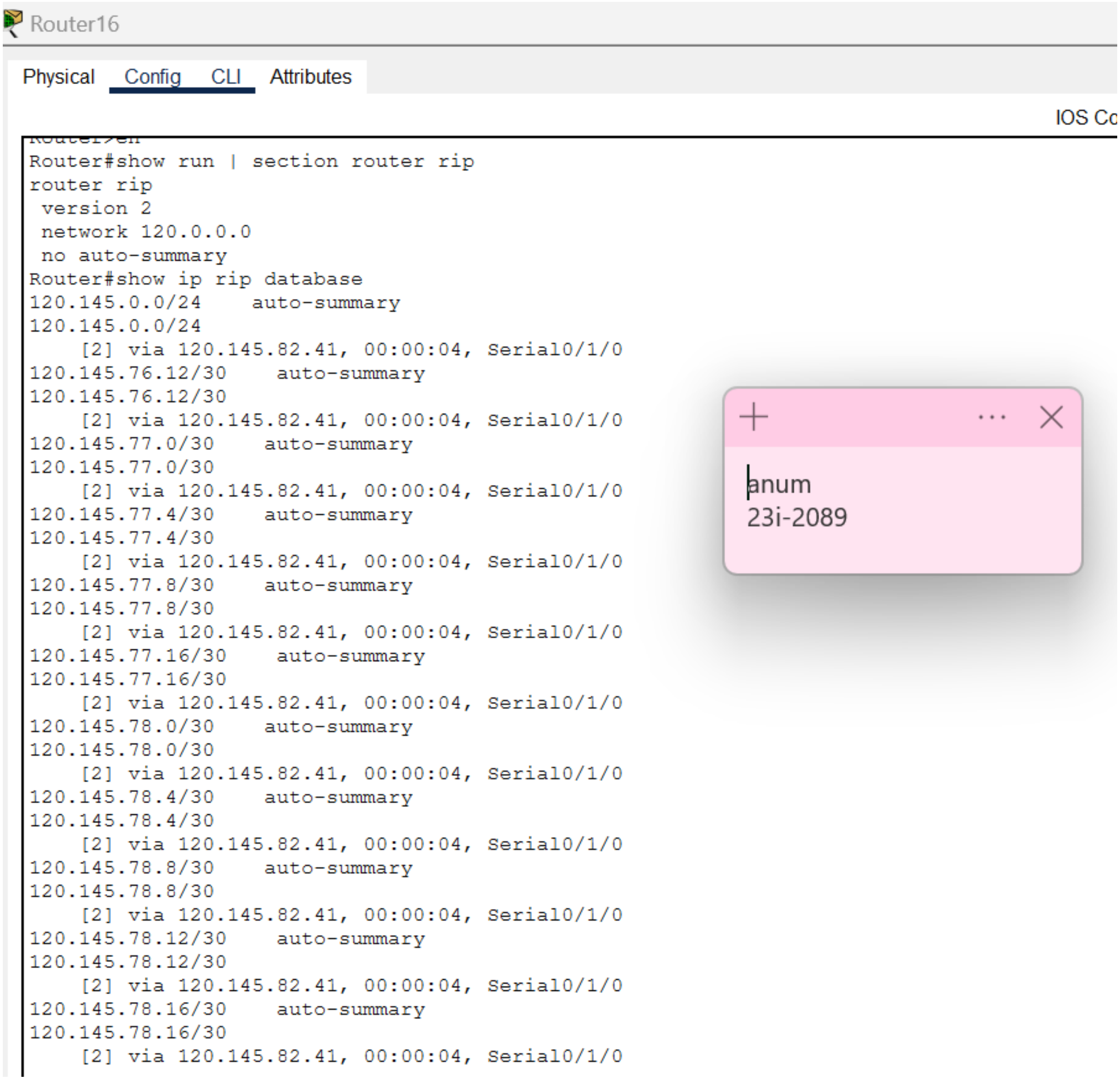
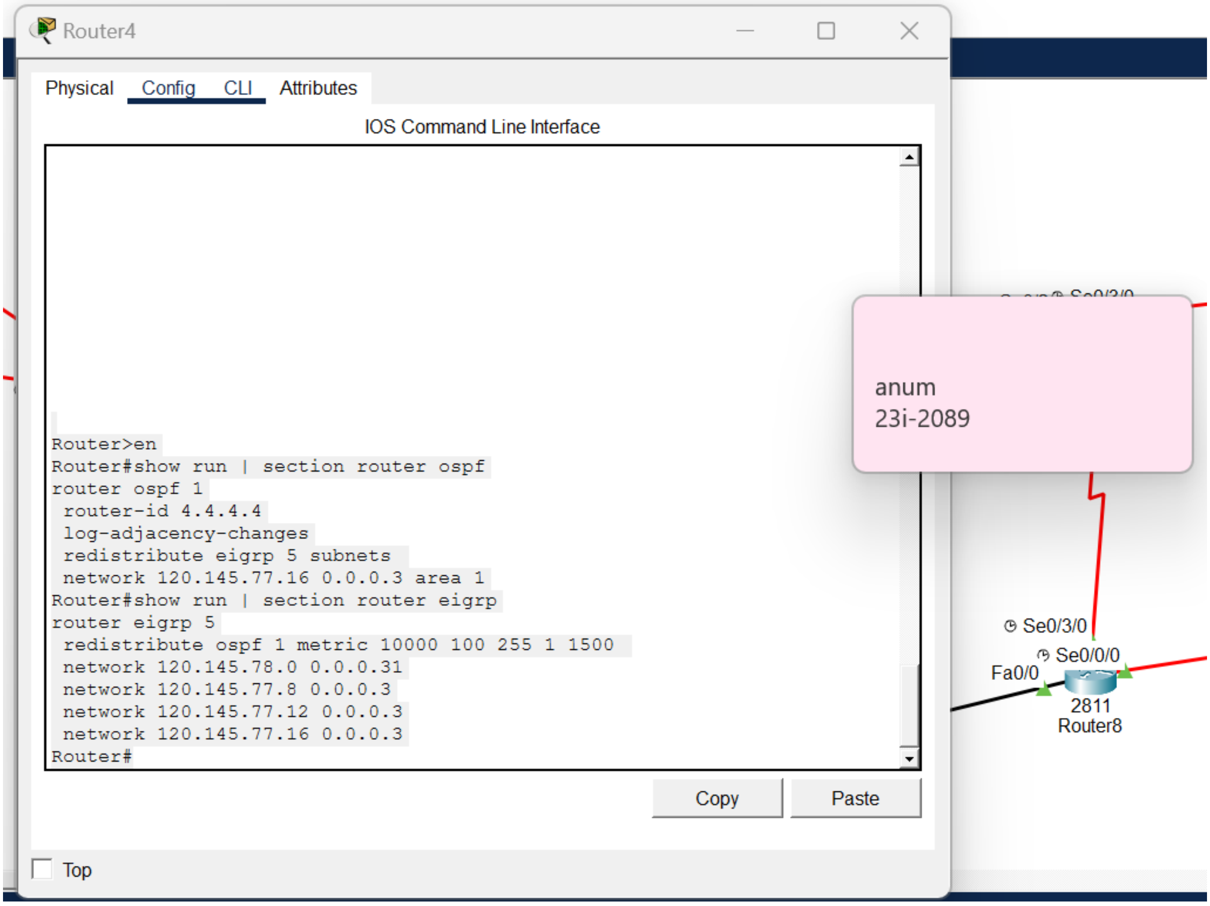
Se0/2/0
Se0/2/0

C. On Router16 for RIP block



2. Redistribution Of Routing Between Areas (OSPF -> EIGRP5 -> RIP)

2.1. On Router4, that is the router between OSPF1 and EIGRP5




```
Router>en
Router#show run | section router ospf
router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
 redistribute eigrp 5 subnets
 network 120.145.77.16 0.0.0.3 area 1
Router#show run | section router eigrp
router eigrp 5
 redistribute ospf 1 metric 10000 100 255 1 1500
 network 120.145.78.0 0.0.0.31
 network 120.145.77.8 0.0.0.3
 network 120.145.77.12 0.0.0.3
 network 120.145.77.16 0.0.0.3
Router#
```

2.2. At Router6 and Router7, EIGRP routes + D EX routes for OSPF networks (like 120.146.0.0/24) is displayed.

Router6:

```
Router>en
Router#show ip route | include D|D EX
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       * - candidate default, U - per-user static route, o - ODR
D EX    120.145.0.0/24 [170/2707456] via 120.145.78.5, 00:39:46, Serial0/2/0
D       120.145.76.12/30 [90/3705856] via 120.145.78.5, 00:39:59, Serial0/2/0
D EX    120.145.77.0/30 [170/2707456] via 120.145.78.5, 00:39:46, Serial0/2/0
D EX    120.145.77.4/30 [170/2707456] via 120.145.78.5, 00:39:46, Serial0/2/0
D       120.145.77.8/30 [90/3705856] via 120.145.78.5, 00:39:58, Serial0/2/0
D       120.145.77.16/30 [90/3193856] via 120.145.78.5, 00:39:58, Serial0/2/0
D       120.145.78.0/30 [90/2681856] via 120.145.78.5, 00:39:59, Serial0/2/0
D       120.145.78.16/30 [90/2681856] via 120.145.78.14, 00:40:00, Serial0/0/0
D       120.145.78.20/30 [90/2681856] via 120.145.78.10, 00:39:59, Serial0/3/0
D       120.145.78.24/30 [90/2681856] via 120.145.78.10, 00:39:59, Serial0/3/0
D       120.145.78.28/30 [90/3193856] via 120.145.78.10, 00:39:59, Serial0/3/0
D       120.145.79.0/27 [90/2172416] via 120.145.78.10, 00:39:59, Serial0/3/0
D       120.145.79.32/27 [90/2172416] via 120.145.78.14, 00:40:00, Serial0/0/0
D       120.145.79.64/27 [90/2684416] via 120.145.78.10, 00:39:59, Serial0/3/0
D EX    120.145.80.0/27 [170/2707456] via 120.145.78.10, 00:39:12, Serial0/3/0
D EX    120.145.80.32/27 [170/2707456] via 120.145.78.10, 00:39:11, Serial0/3/0
D EX    120.145.80.64/27 [170/2707456] via 120.145.78.10, 00:39:11, Serial0/3/0
D EX    120.145.81.0/30 [170/2707456] via 120.145.78.10, 00:39:59, Serial0/3/0
D EX    120.145.81.4/30 [170/2707456] via 120.145.78.10, 00:39:12, Serial0/3/0
D EX    120.145.81.8/30 [170/2707456] via 120.145.78.10, 00:39:12, Serial0/3/0

Router#
```

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Router7:

```
Router>en
Router#show ip route | include D|D EX
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       * - candidate default, U - per-user static route, o - ODR
D EX    120.145.0.0/24 [170/3219456] via 120.145.78.13, 00:58:14, Serial0/3/0
D       120.145.76.12/30 [90/4217856] via 120.145.78.13, 00:58:27, Serial0/3/0
D EX    120.145.77.0/30 [170/3219456] via 120.145.78.13, 00:58:14, Serial0/3/0
D EX    120.145.77.4/30 [170/3219456] via 120.145.78.13, 00:58:14, Serial0/3/0
D       120.145.77.8/30 [90/4217856] via 120.145.78.13, 00:58:26, Serial0/3/0
D       120.145.77.16/30 [90/3705856] via 120.145.78.13, 00:58:26, Serial0/3/0
D       120.145.78.0/30 [90/3193856] via 120.145.78.13, 00:58:27, Serial0/3/0
D       120.145.78.4/30 [90/2681856] via 120.145.78.13, 00:58:27, Serial0/3/0
D       120.145.78.8/30 [90/2681856] via 120.145.78.13, 00:58:28, Serial0/3/0
D       120.145.78.20/30 [90/2681856] via 120.145.78.18, 00:58:27, Serial0/0/0
D       120.145.78.24/30 [90/3193856] via 120.145.78.13, 00:58:27, Serial0/3/0
D       120.145.78.28/30 [90/2681856] via 120.145.78.18, 00:58:27, Serial0/0/0
D       120.145.79.0/27 [90/2684416] via 120.145.78.13, 00:58:27, Serial0/3/0
D       120.145.79.64/27 [90/2172416] via 120.145.78.18, 00:58:27, Serial0/0/0
D EX    120.145.80.0/27 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0
D EX    120.145.80.32/27 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0
D EX    120.145.80.64/27 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0
D EX    120.145.81.0/30 [170/2707456] via 120.145.78.18, 00:58:27, Serial0/0/0
D EX    120.145.81.4/30 [170/2707456] via 120.145.78.18, 00:57:40, Serial0/0/0
D EX    120.145.81.8/30 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0

Router#
```

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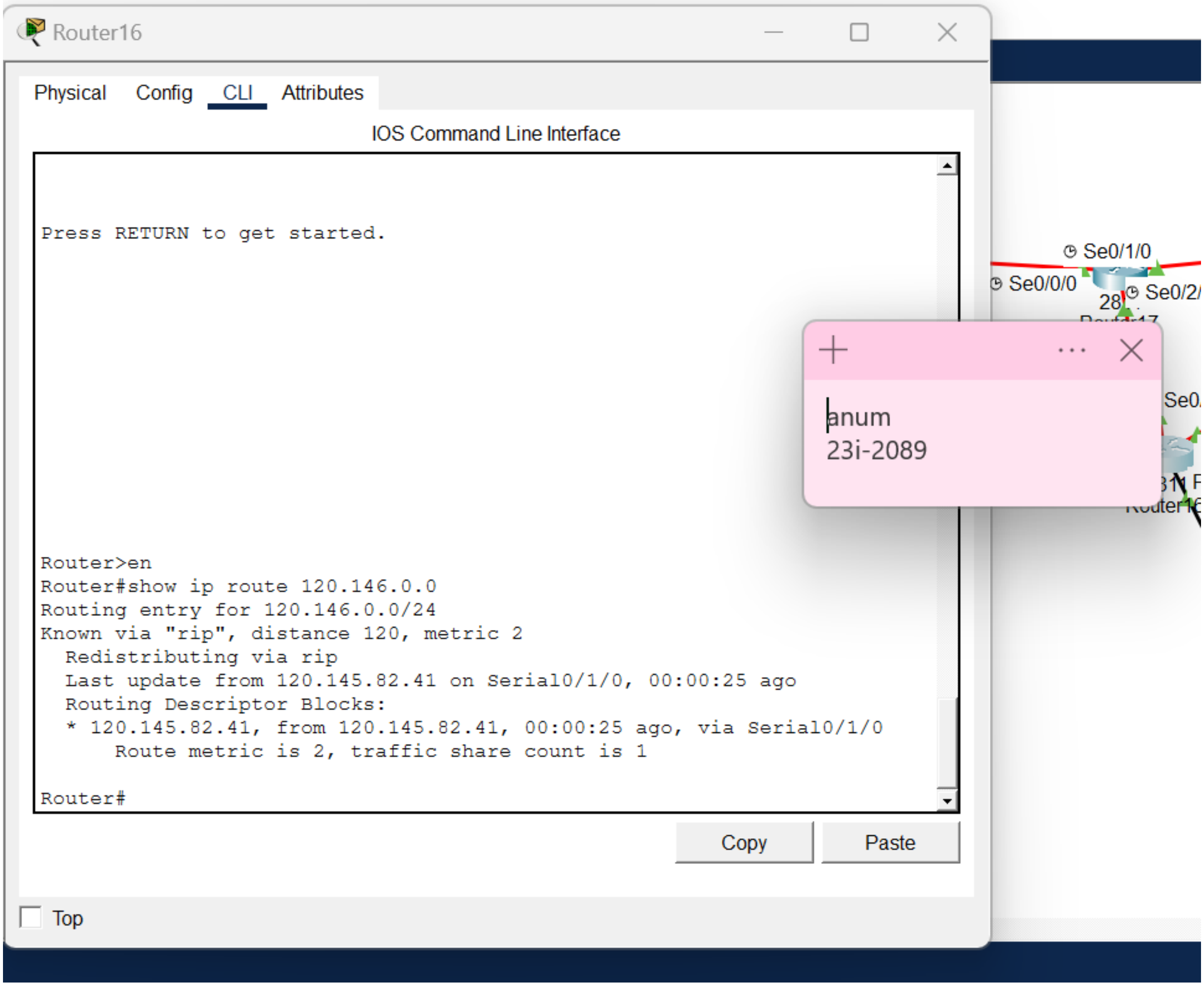
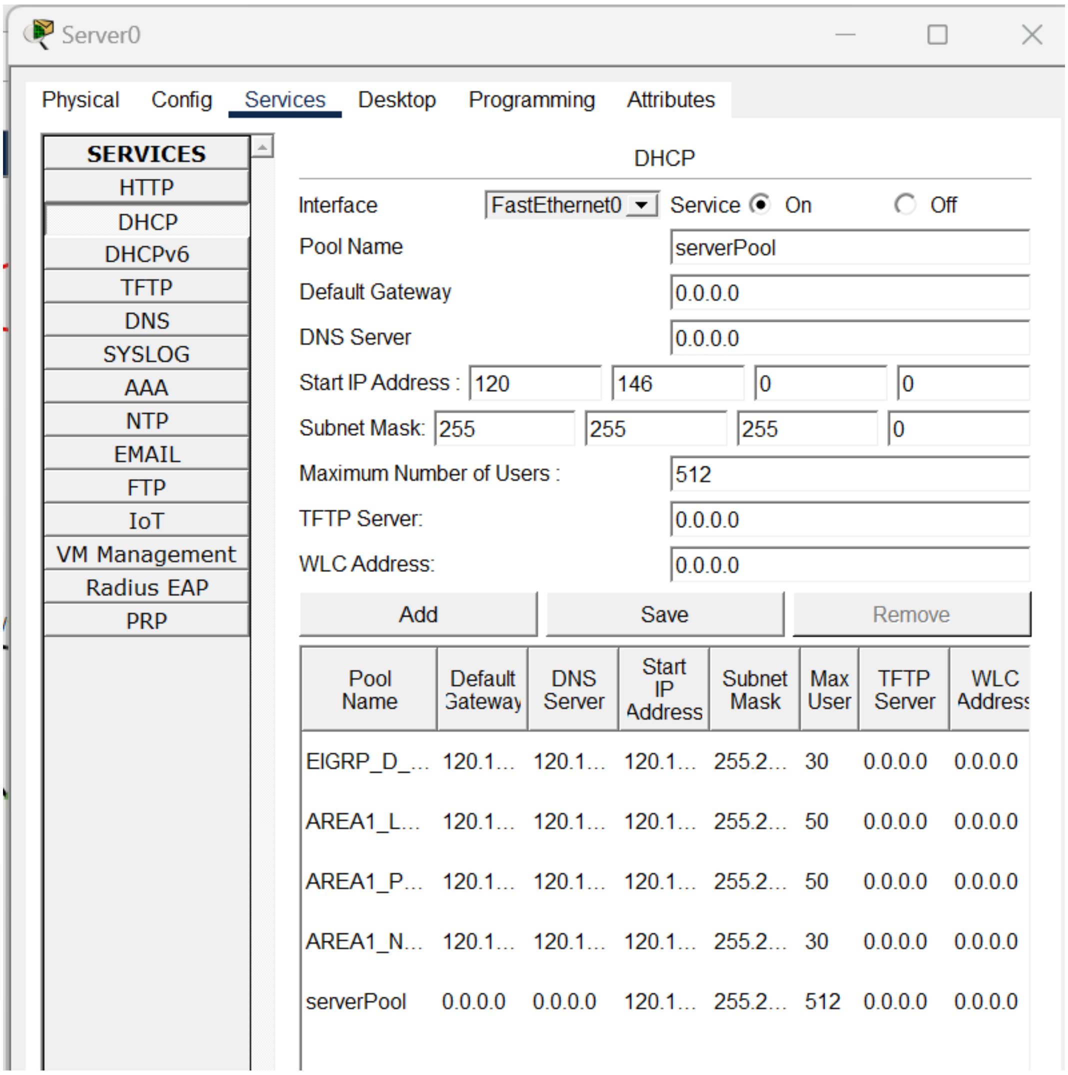
2.3. At router16 for Rip area



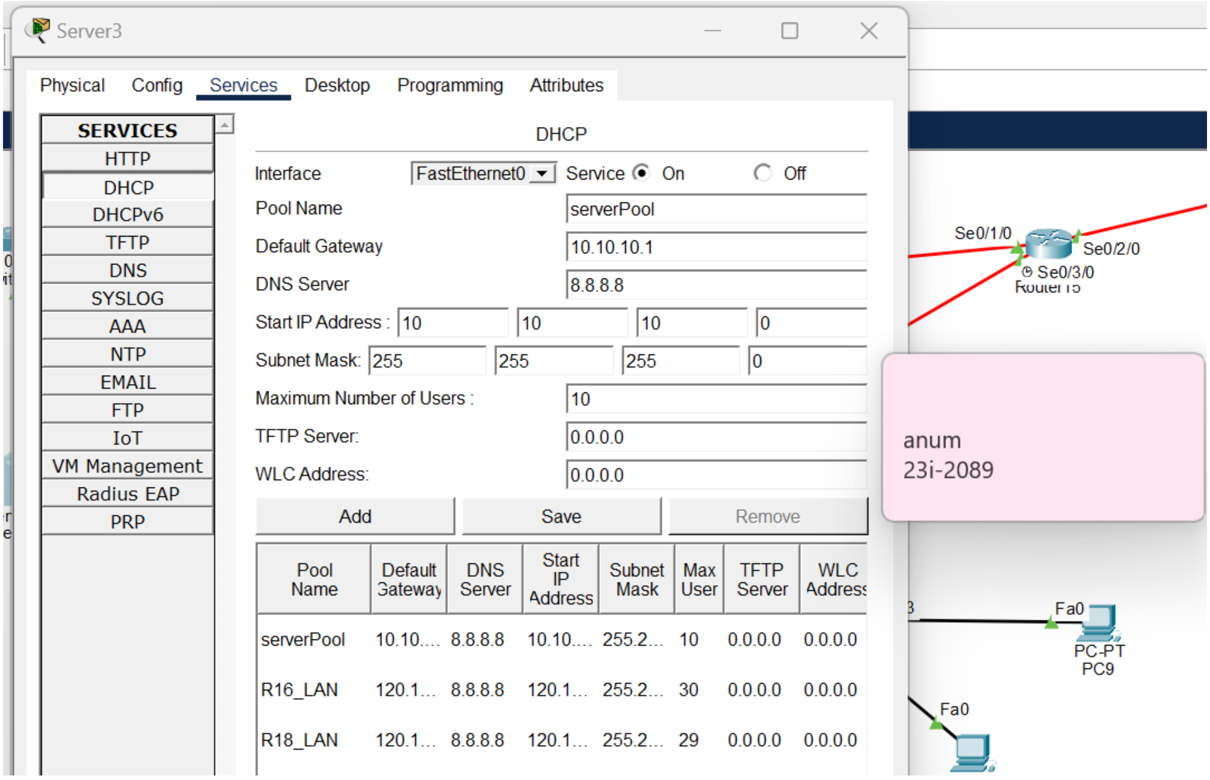
3. DHCP in OSPF, EIGRP and RIP

DHCP pools in server3 (RIP) and server0 (OSPF1)

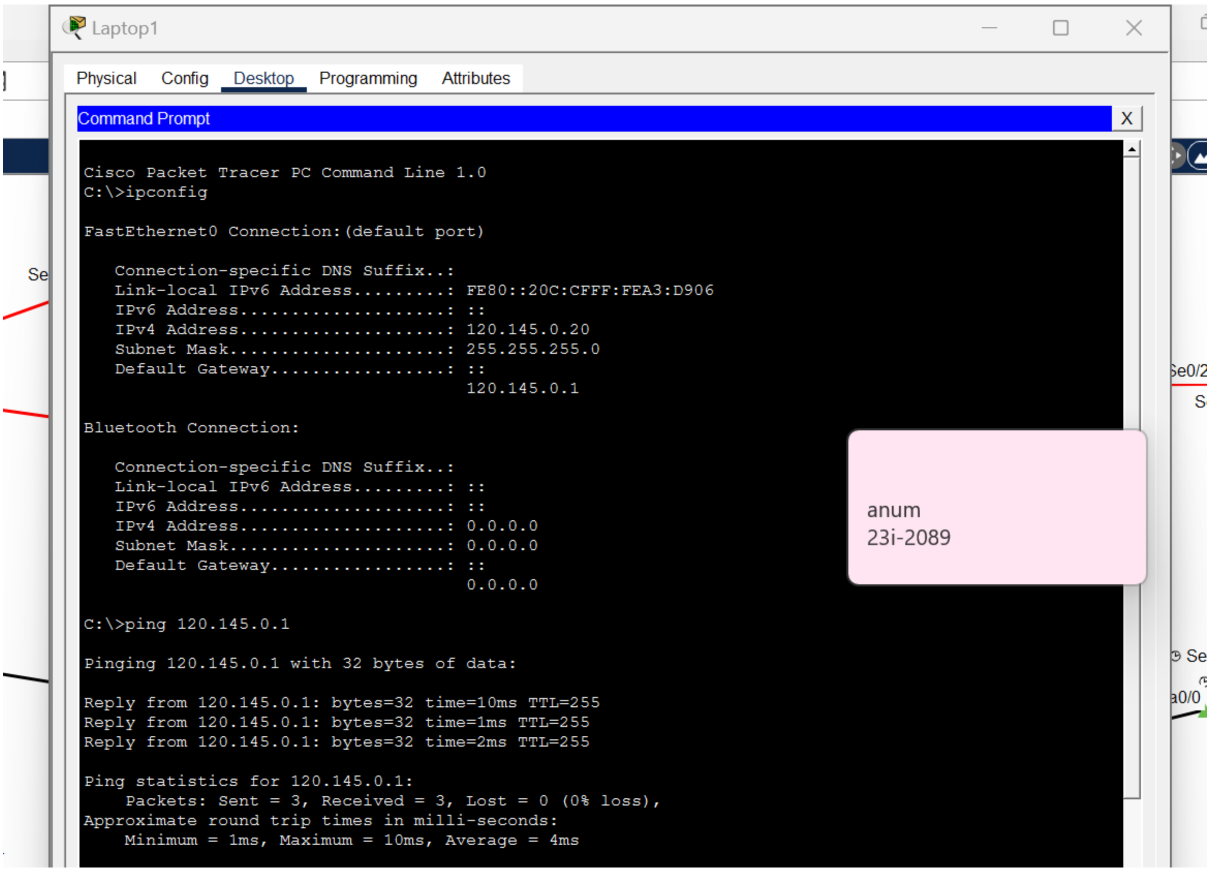
Server0:

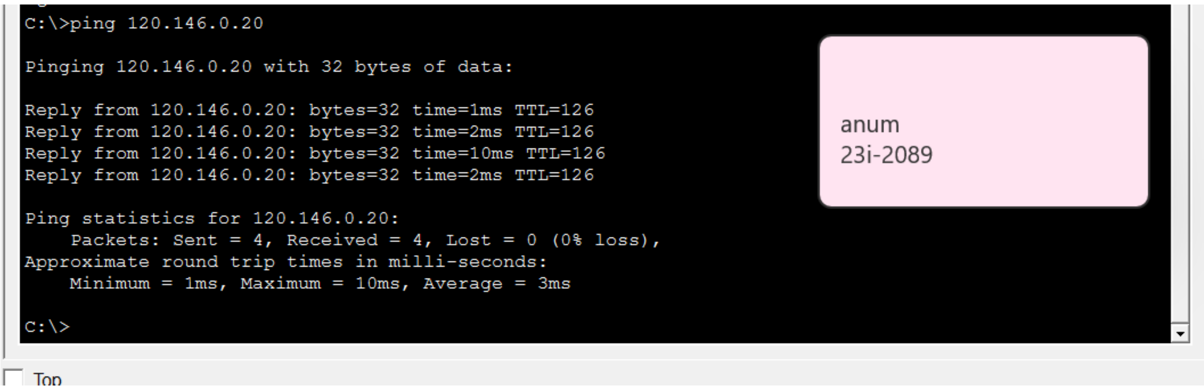


Server3:



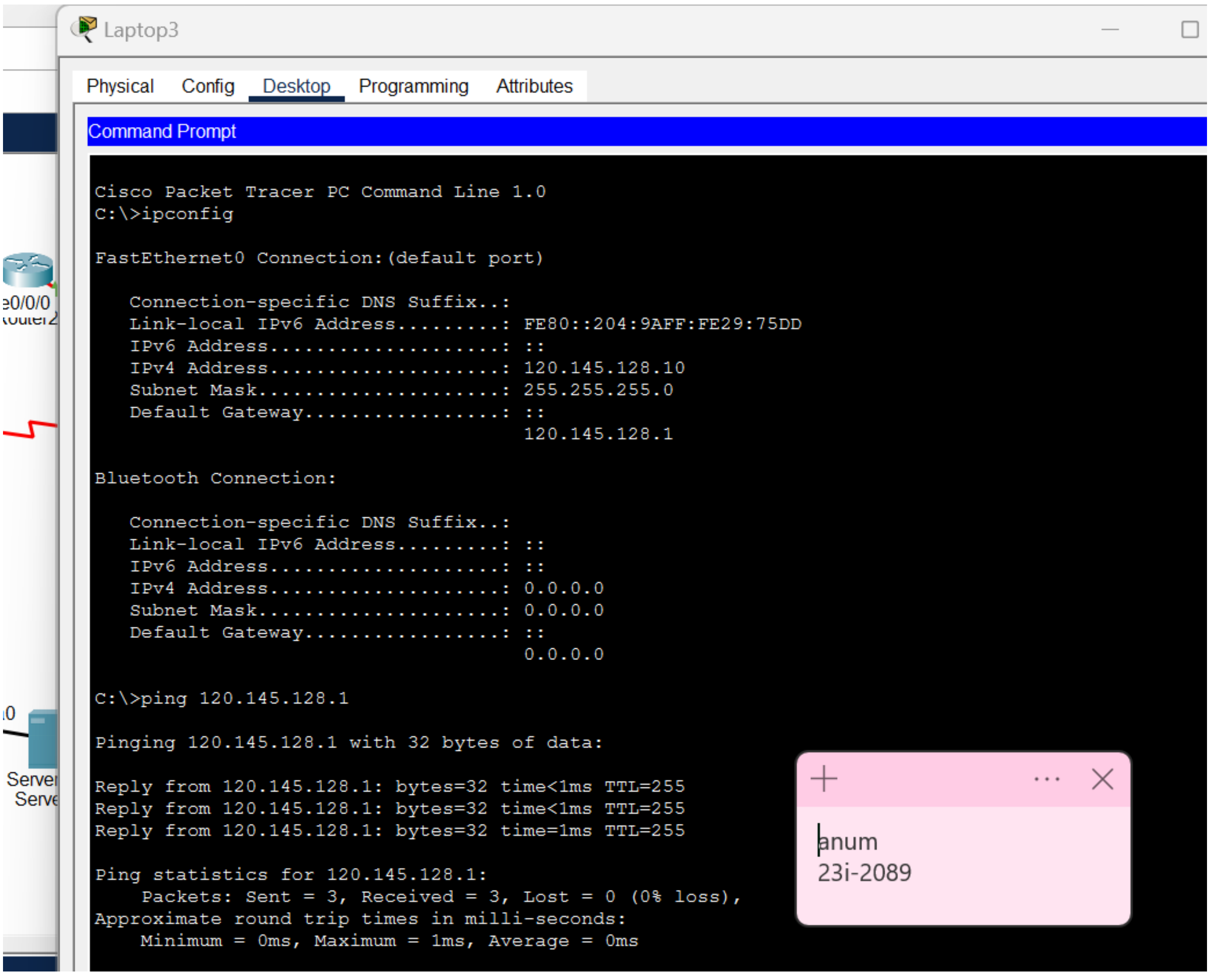
3.1. at laptop1 (network A) of OSPF





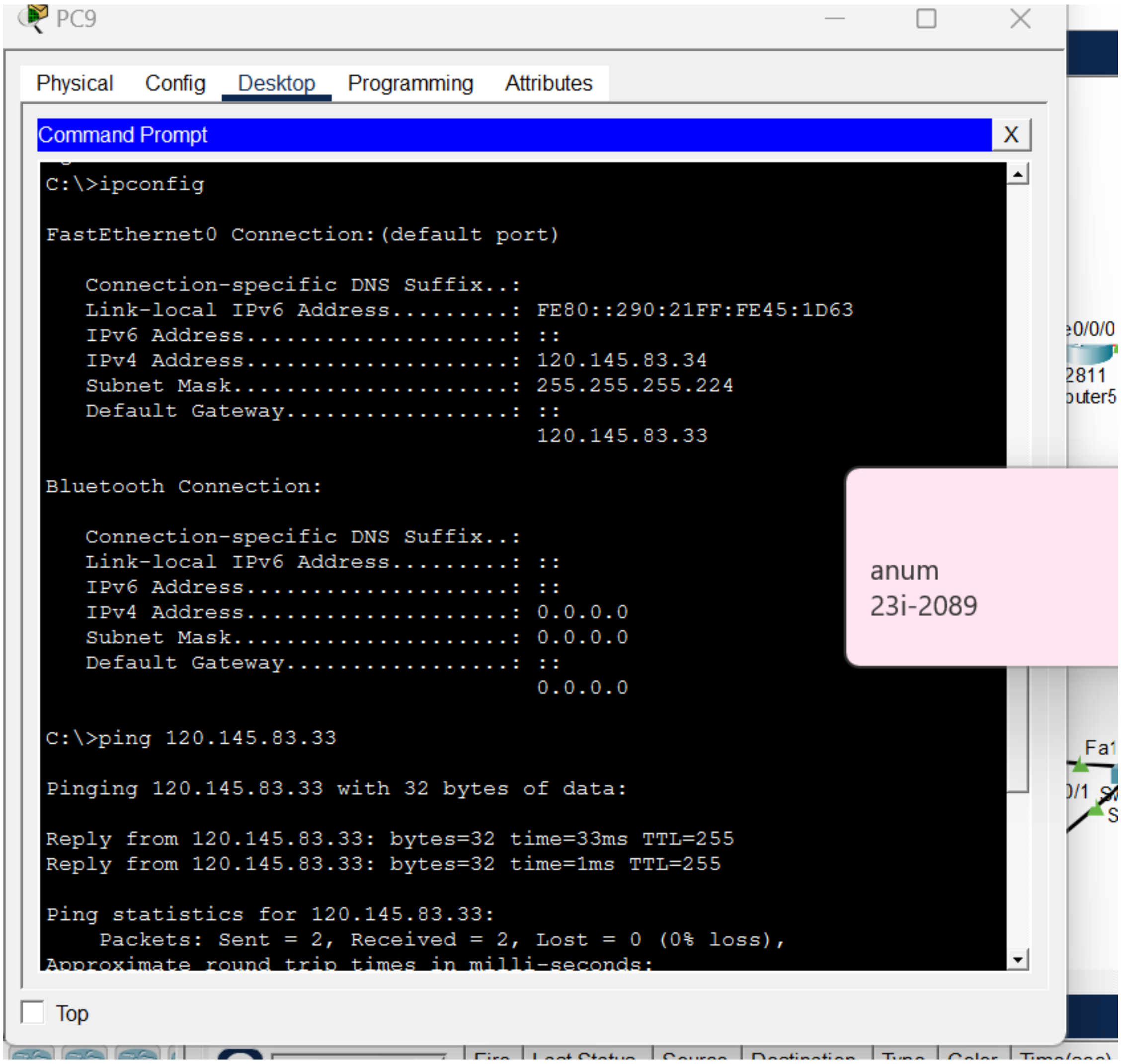
3.2. At Network B of OSFP.

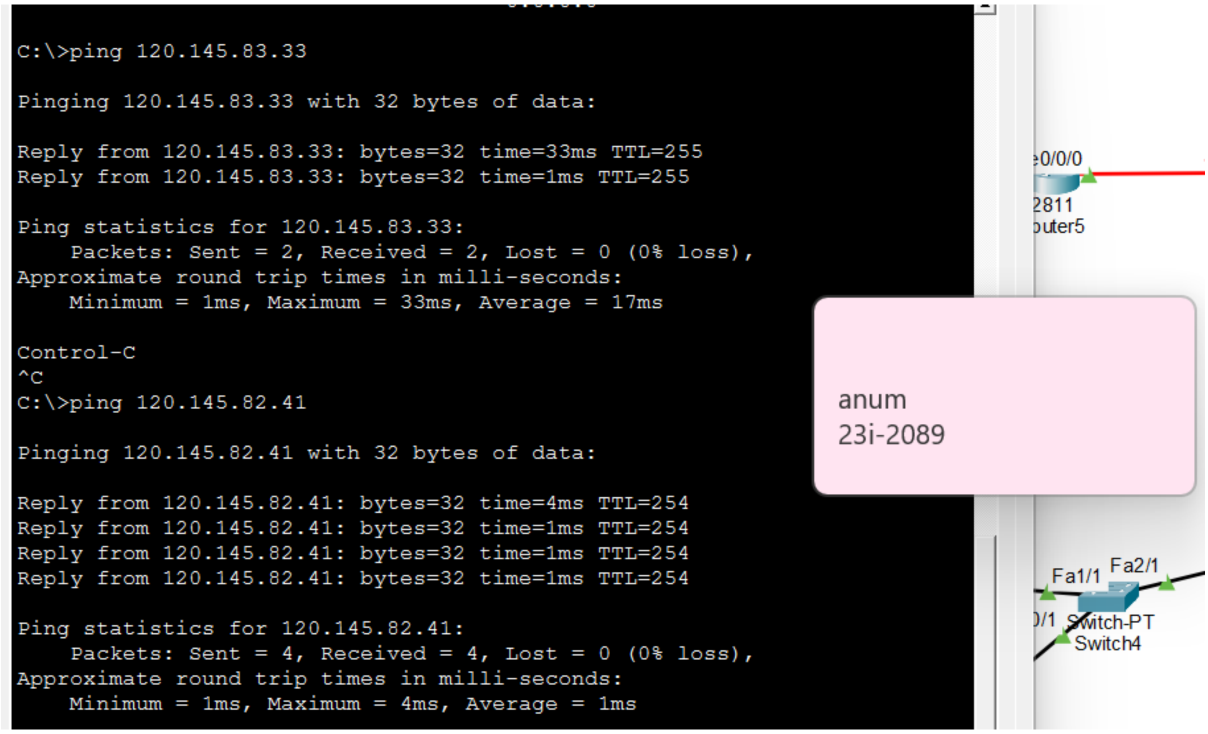
Laptop3:



3.3. At Network C OSPF1

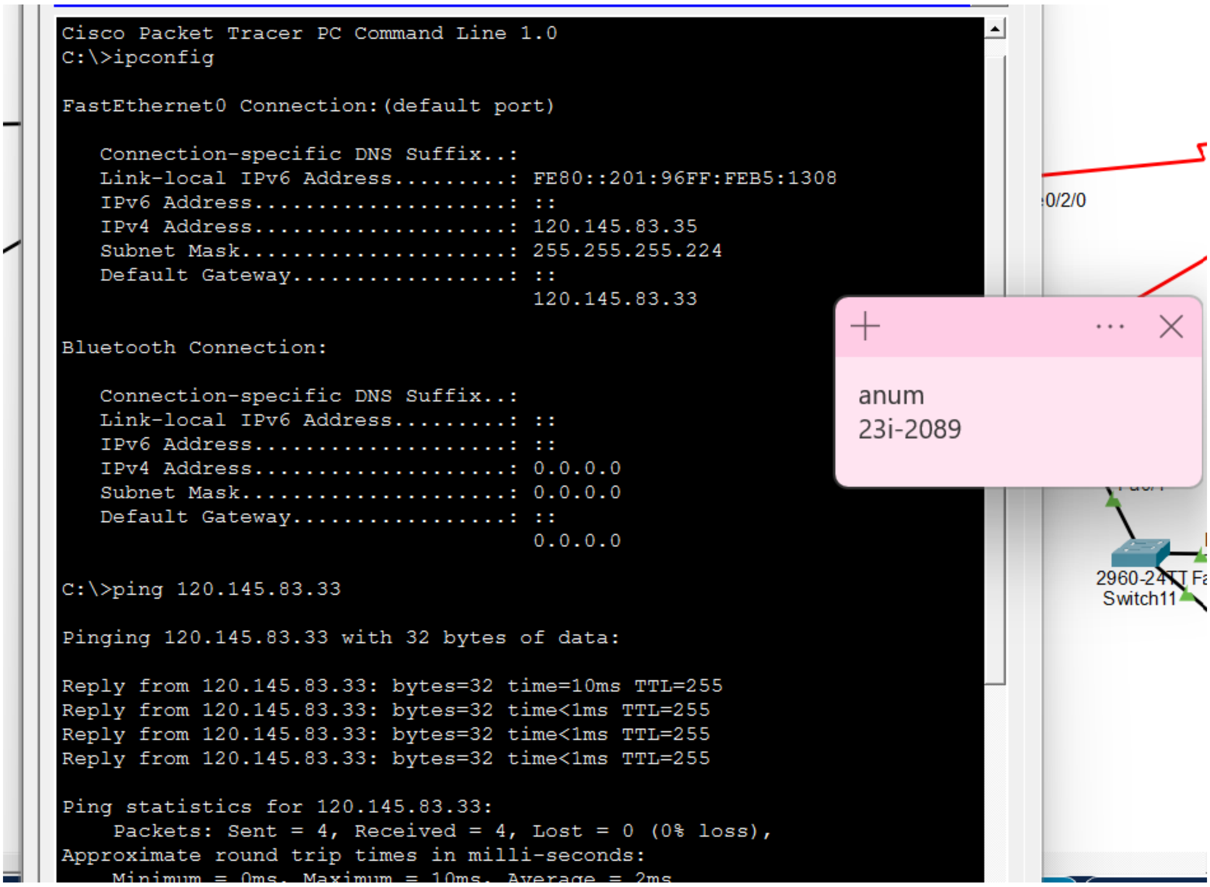
3.4. Network D in EIGRP5 Block





```
C:\>ping 120.145.83.33

Pinging 120.145.83.33 with 32 bytes of data:

Reply from 120.145.83.33: bytes=32 time=33ms TTL=255
Reply from 120.145.83.33: bytes=32 time=1ms TTL=255

Ping statistics for 120.145.83.33:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 33ms, Average = 17ms

Control-C
^C
C:\>ping 120.145.82.41

Pinging 120.145.82.41 with 32 bytes of data:

Reply from 120.145.82.41: bytes=32 time=4ms TTL=254
Reply from 120.145.82.41: bytes=32 time=1ms TTL=254
Reply from 120.145.82.41: bytes=32 time=1ms TTL=254
Reply from 120.145.82.41: bytes=32 time=1ms TTL=254

Ping statistics for 120.145.82.41:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 1ms
```

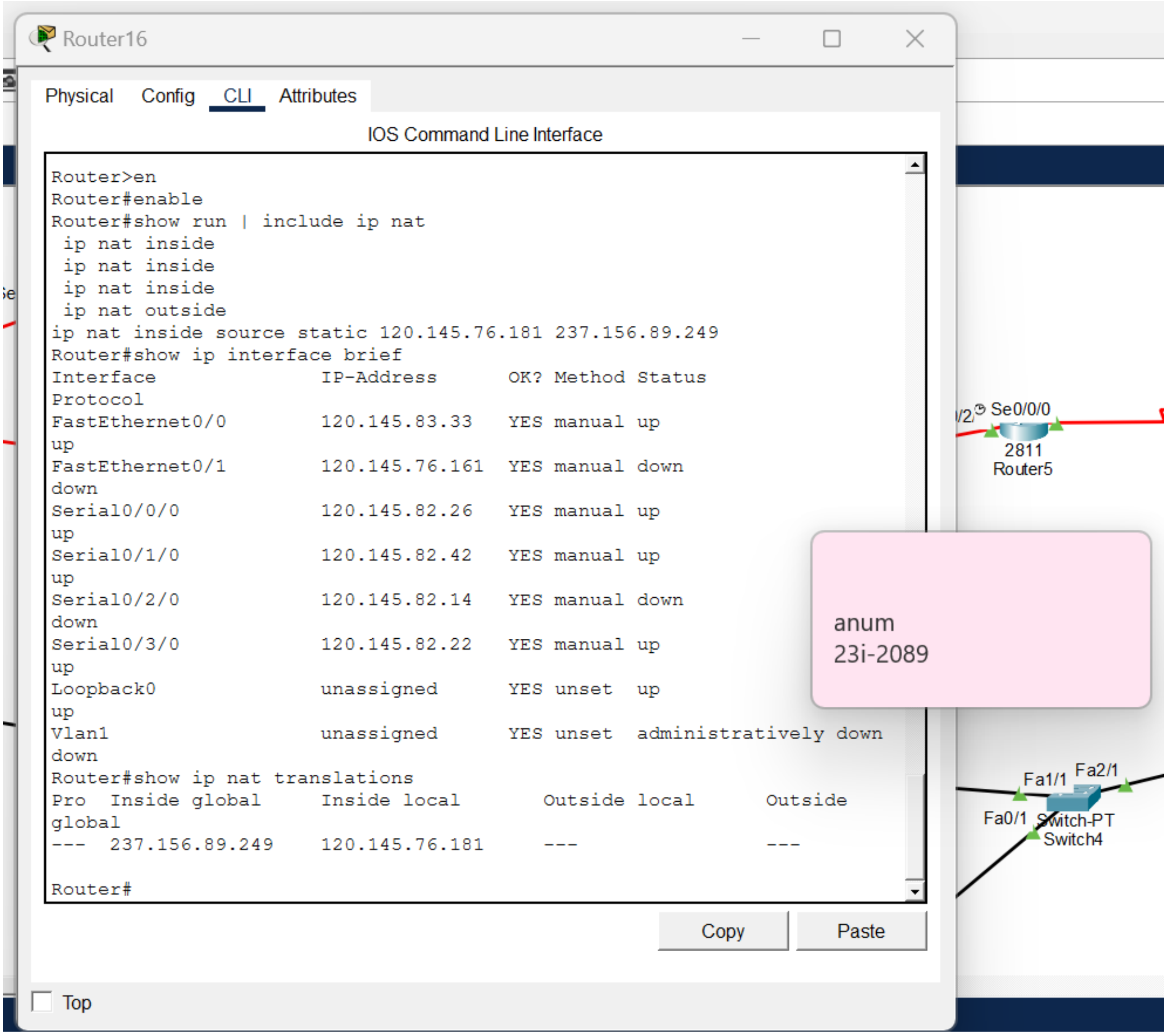

3.5. Network J in RIP



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:96FF:FEB5:1308
   IPv6 Address....................: ::
   IPv4 Address....................: 120.145.83.35
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     120.145.83.33

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 120.145.83.33

Pinging 120.145.83.33 with 32 bytes of data:

Reply from 120.145.83.33: bytes=32 time=10ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255

Ping statistics for 120.145.83.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

4. NAT Verification

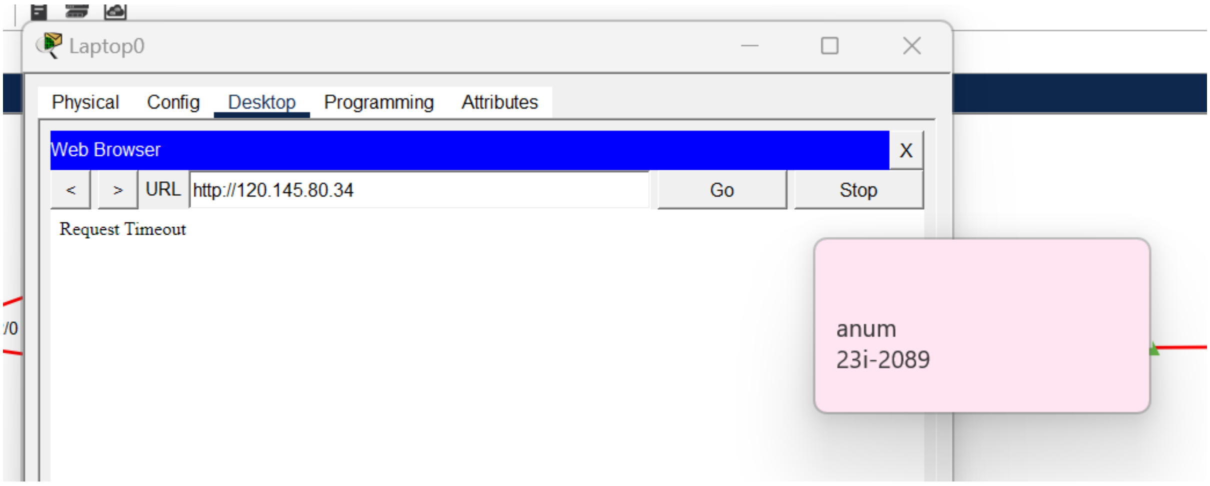
4.1. Configuration and Mapping



5. ACL Behavior and Verification

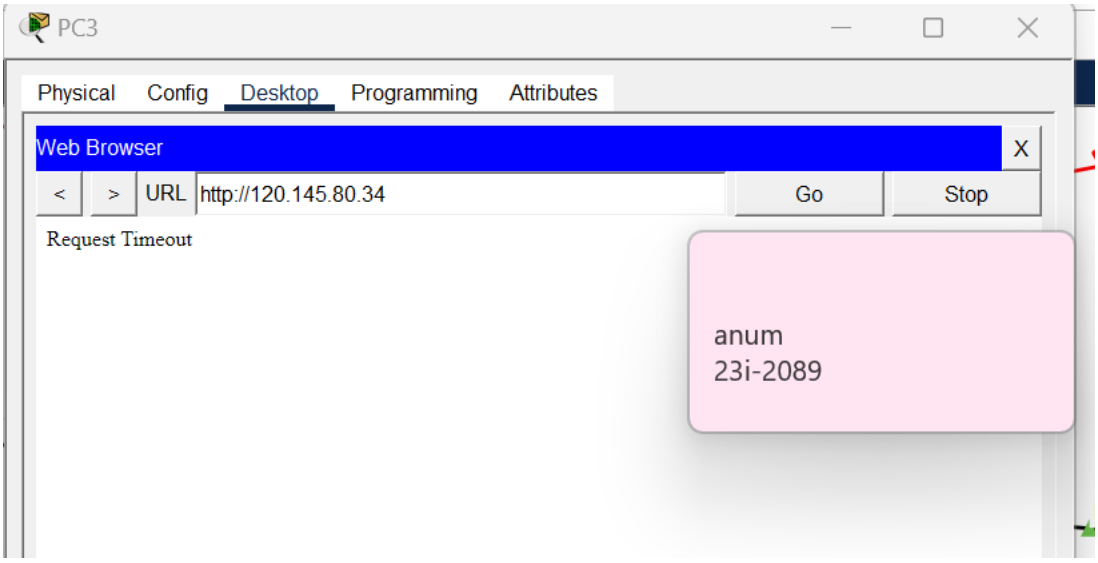
5.1. At network A

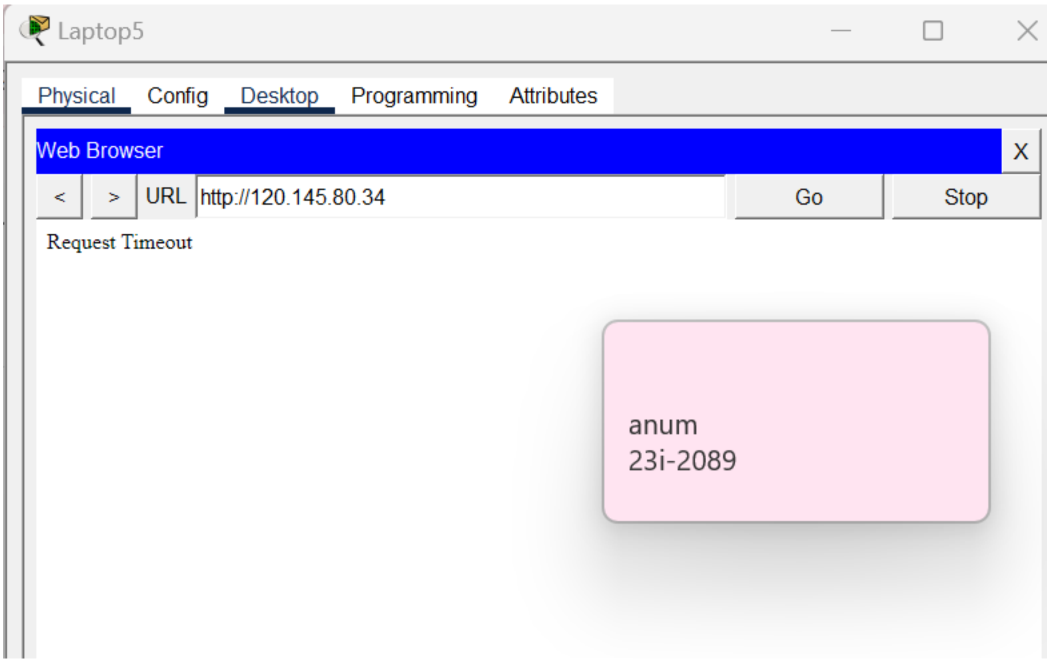
Laptop0 in Network A is blocked from HTTP by using http://120.145.80.34.



5.2. At network D

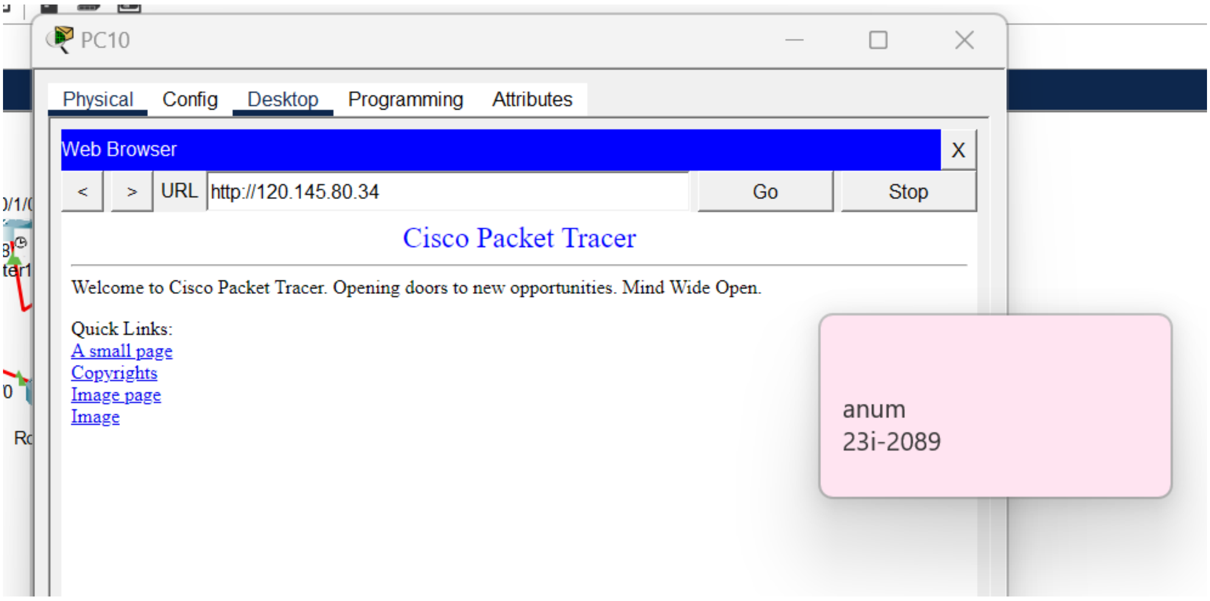
Network D consists of PC3 and Laptop5. Hence the network D is blocked in this case.





5.3. Case of Non-Block

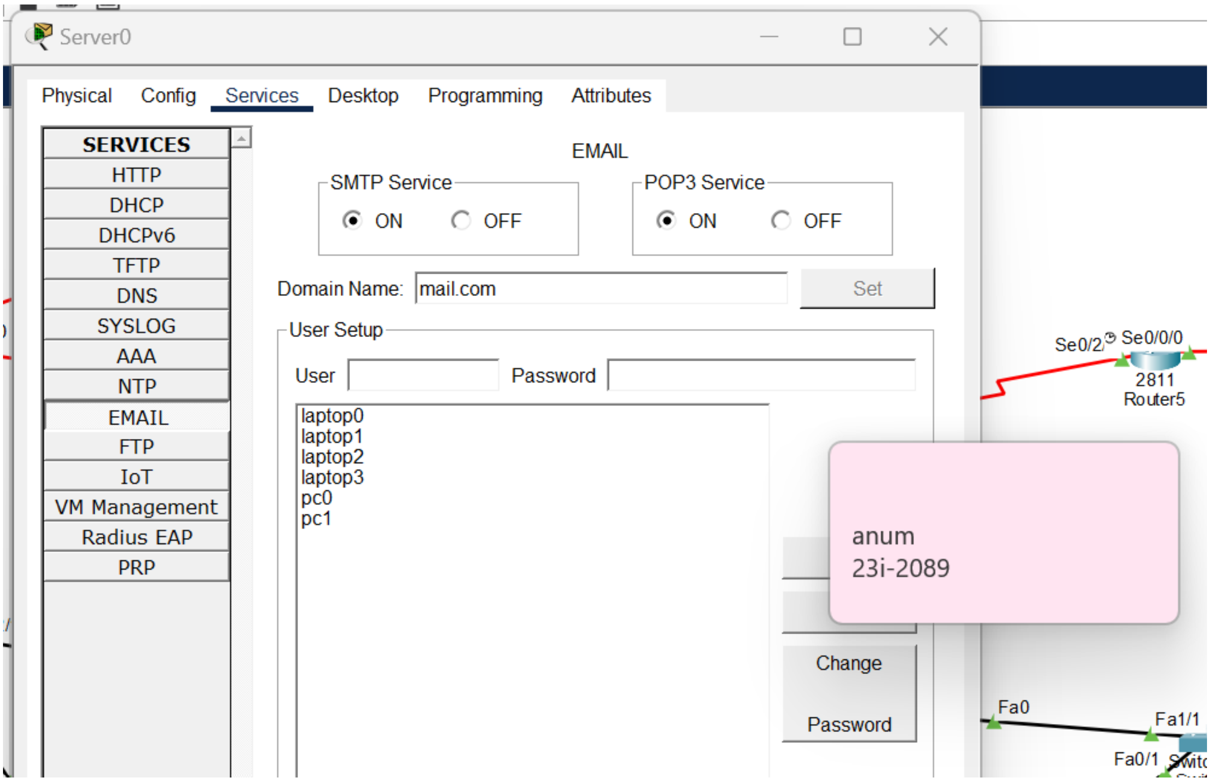
For instance, Pc10 in network J isn't blocked from HTTP. Hence;

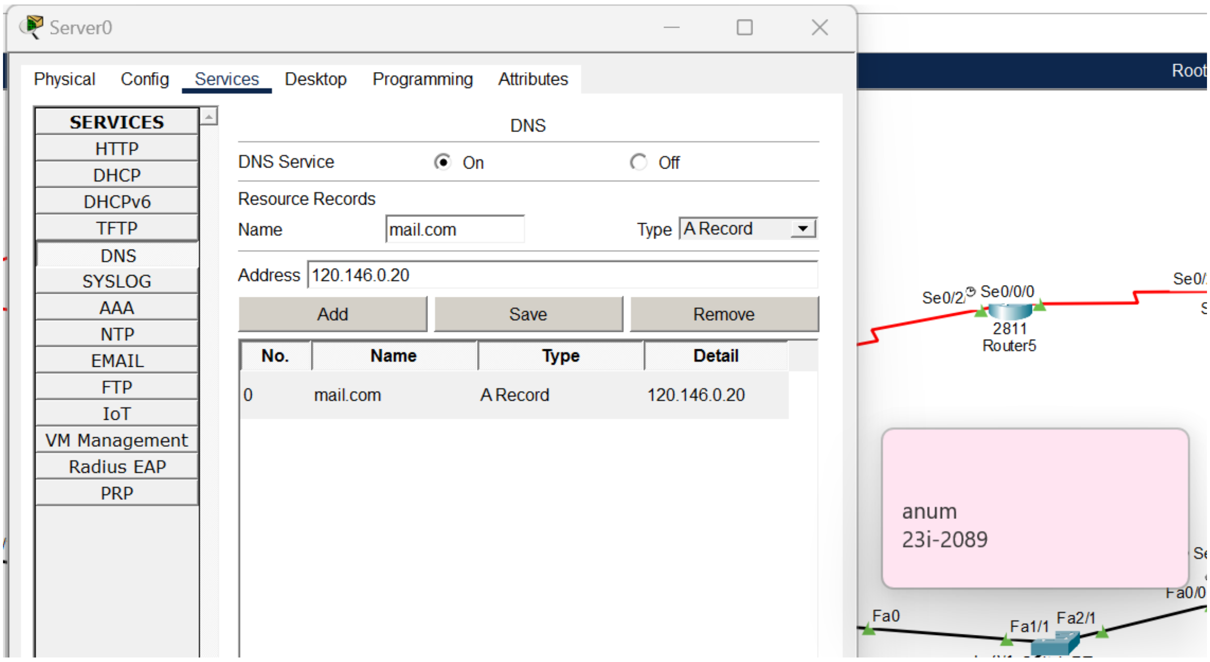


6. EMAIL SMTP

6.1. Confirming Mail Services.

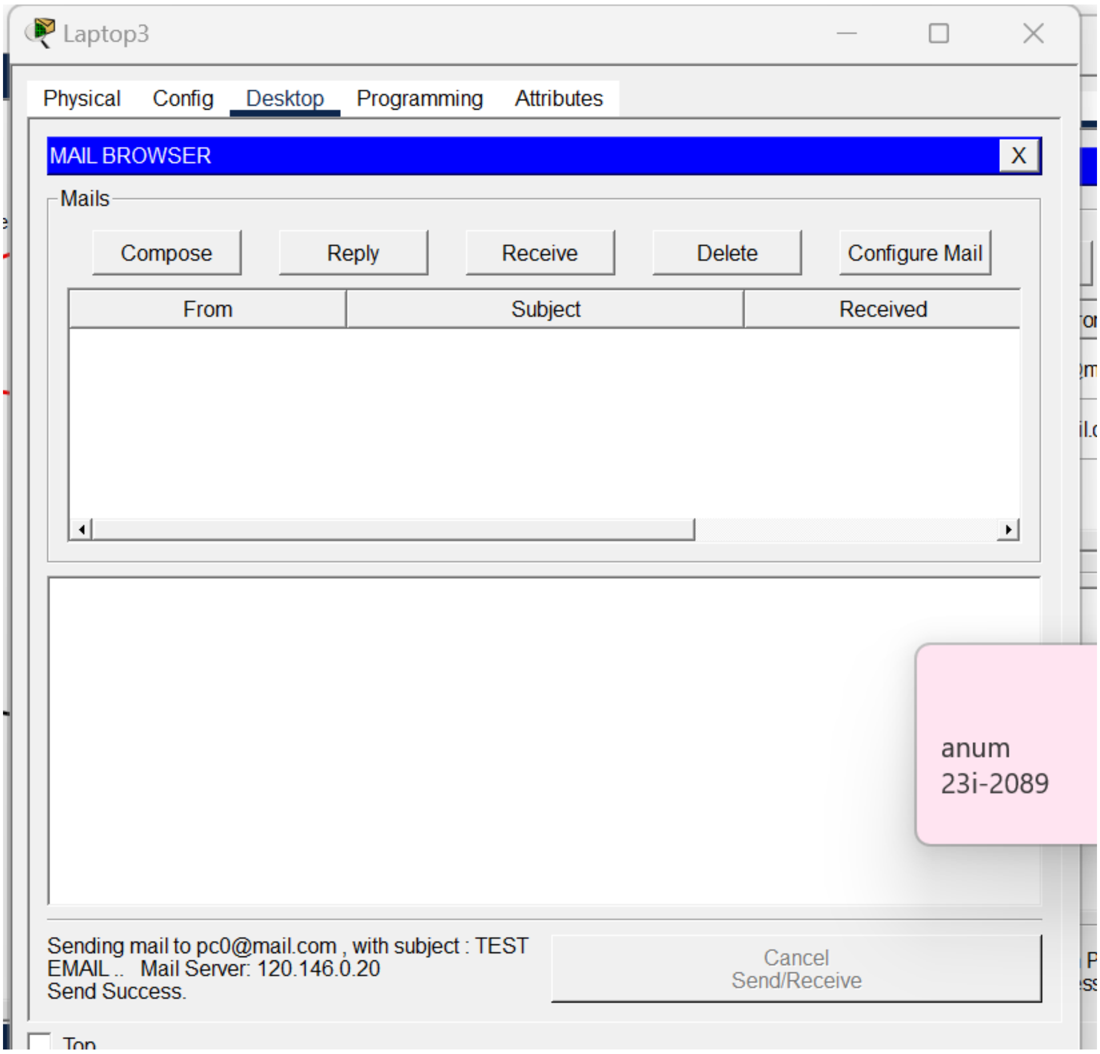
DNS and SMTP were turned on and assigned with domain names, usernames and passwords.



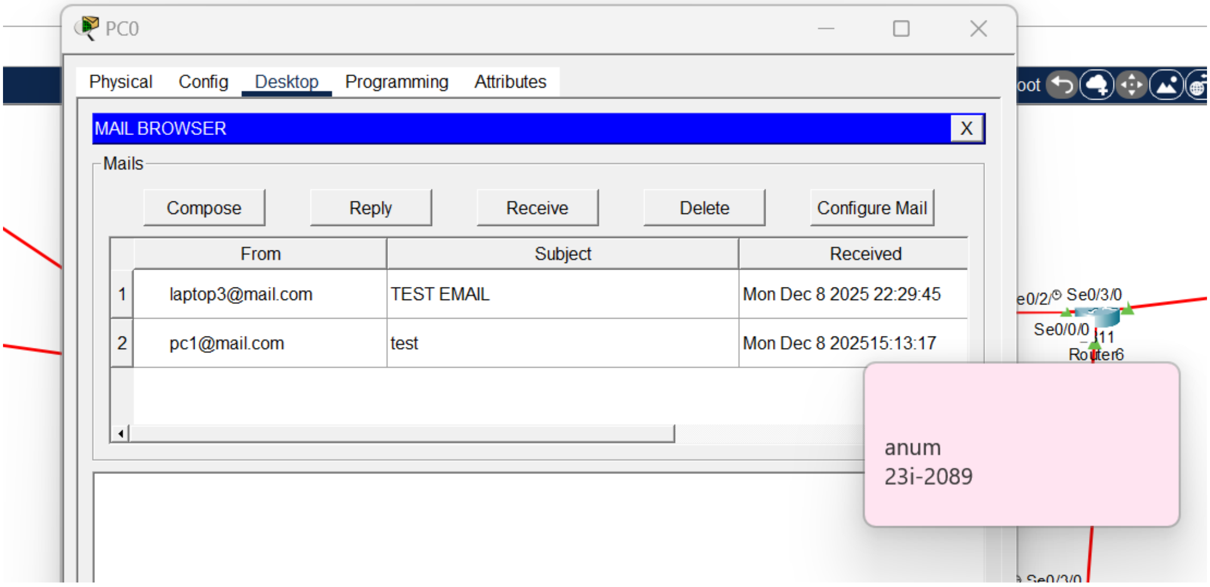


6.2. Testing and Verification

In network A, mail was sent by laptop3 to Pc0.

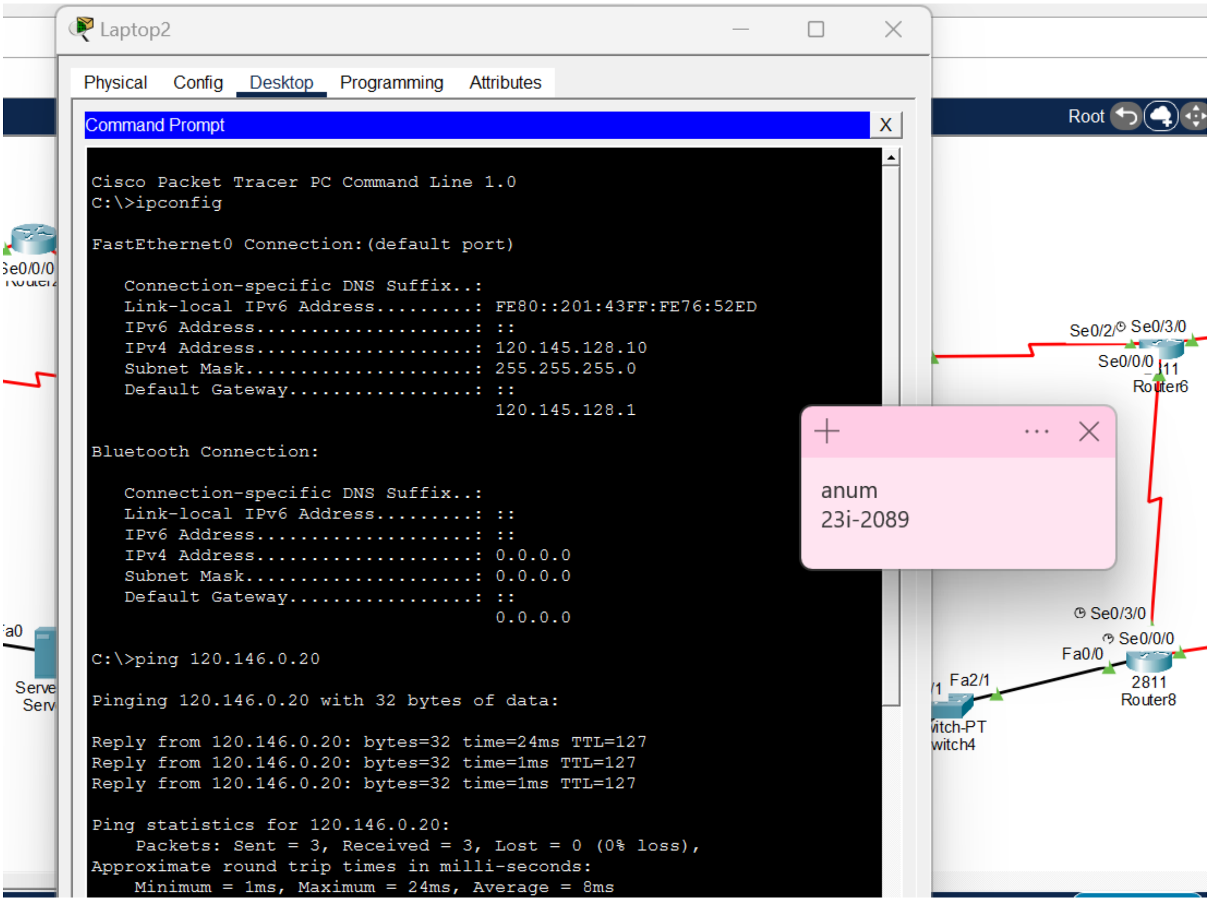


Here, Pc0 has received mail from laptop3.

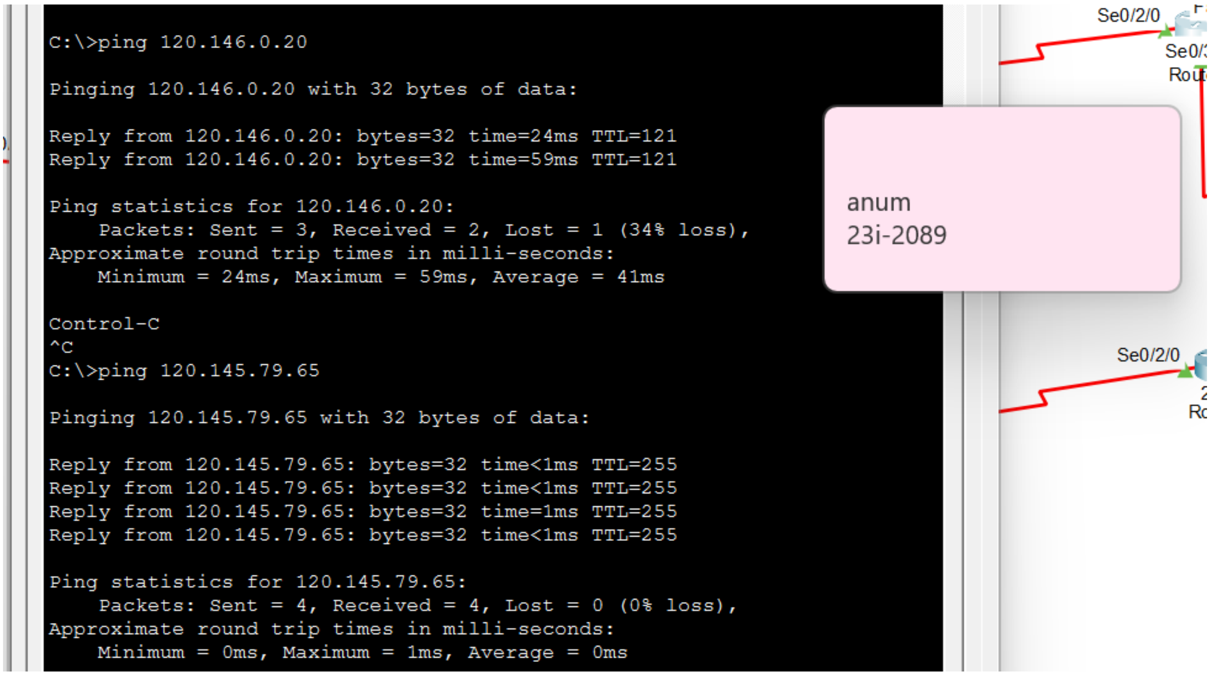


7. Final Connectivity Check

7.1. For Network B of OSPF1

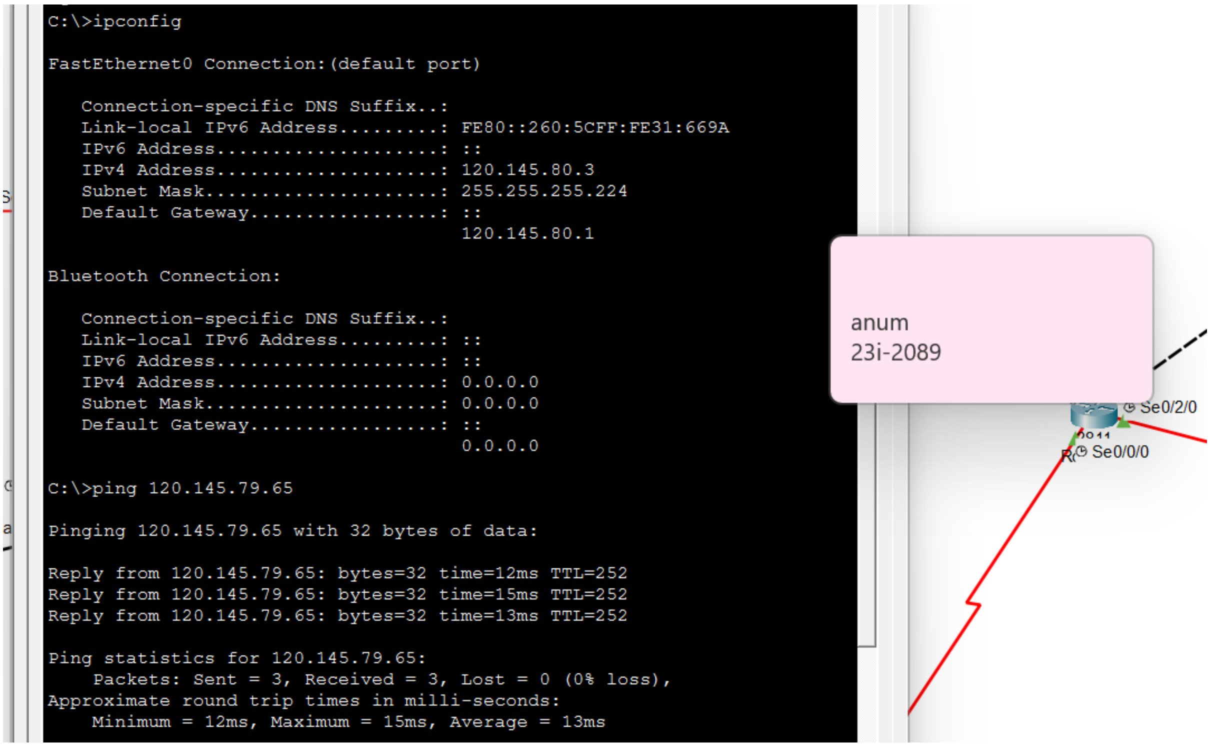


Laptop6 in Network E obtained IP 120.145.79.66/27 via DHCP with default gateway 120.145.79.65. Successful ping to the gateway confirms local connectivity, and successful ping to 120.146.0.20 did not happen because network E wasn't assigned a DHCP pool.



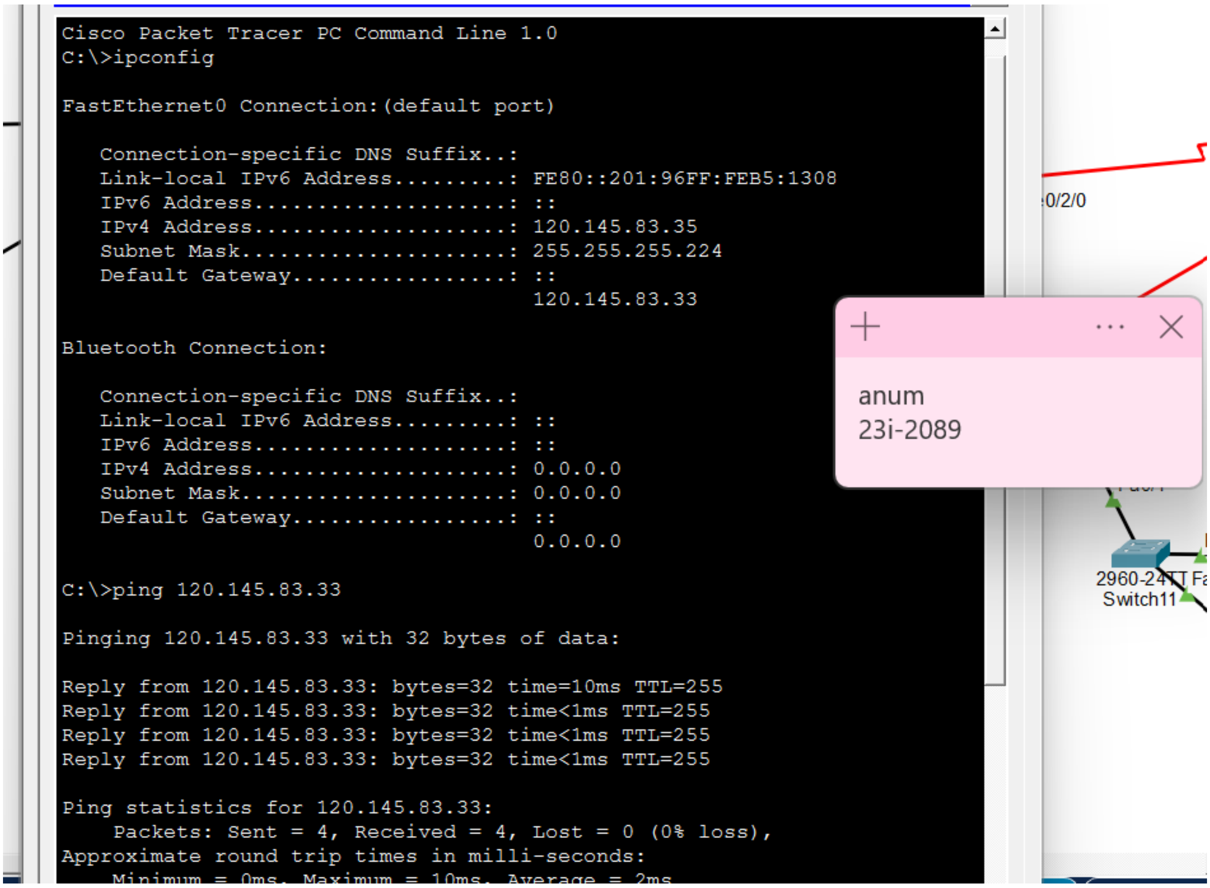
```
C:\>ping 120.146.0.20

Pinging 120.146.0.20 with 32 bytes of data:

Reply from 120.146.0.20: bytes=32 time=24ms TTL=121
Reply from 120.146.0.20: bytes=32 time=59ms TTL=121

Ping statistics for 120.146.0.20:
    Packets: Sent = 3, Received = 2, Lost = 1 (34% loss),
Approximate round trip times in milli-seconds:
    Minimum = 24ms, Maximum = 59ms, Average = 41ms

Control-C
^C
C:\>ping 120.145.79.65

Pinging 120.145.79.65 with 32 bytes of data:

Reply from 120.145.79.65: bytes=32 time<1ms TTL=255
Reply from 120.145.79.65: bytes=32 time<1ms TTL=255
Reply from 120.145.79.65: bytes=32 time=1ms TTL=255
Reply from 120.145.79.65: bytes=32 time<1ms TTL=255

Ping statistics for 120.145.79.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```


For connectivity in OSPF2, network G. Hence, Successful ping to the gateway confirms local connectivity.



```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:5CFF:FE31:669A
   IPv6 Address....................: ::
   IPv4 Address....................: 120.145.80.3
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     120.145.80.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 120.145.79.65

Pinging 120.145.79.65 with 32 bytes of data:

Reply from 120.145.79.65: bytes=32 time=12ms TTL=252
Reply from 120.145.79.65: bytes=32 time=15ms TTL=252
Reply from 120.145.79.65: bytes=32 time=13ms TTL=252

Ping statistics for 120.145.79.65:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 15ms, Average = 13ms
```

PC10 can reach its gateway. Hence, DHCP and local connectivity are working for Network J.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:96FF:FEB5:1308
   IPv6 Address....................: ::
   IPv4 Address....................: 120.145.83.35
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     120.145.83.33

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 120.145.83.33

Pinging 120.145.83.33 with 32 bytes of data:

Reply from 120.145.83.33: bytes=32 time=10ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255

Ping statistics for 120.145.83.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```


8. Challenges

- Overlapping in IPs assigned to Networks and in network statements under OSPF/EIGRP.
- Took time to implement redistribution in routers.
- ACL application was causing blockage in both devices of network A.
- Nat was mainly implemented as configuration but couldn't be done as end-to-end for every host.

9. Conclusion

The project was done by using protocols, services and routing. The main areas were OSPF, EIGRP and RIP. The purpose was to interconnect the areas, configuring NAT, selective access to browser via ACL, email in area 1 and DHCP design by creating a topology on Cisco packet tracer.

10. Topology